

Stakeholder Question Catalogue

STAKEHOLDER 19

Waste-management teams

Eighty operational questions connecting proposed privacy-preserving evidence, current authoritative records, explicit analysis, safe answer limits and concrete waste-service decisions.

EVIDENCE-BOUNDED BUILDING DECISION SUPPORT

Sensors, integrations and analytics remain proposed until independently verified. Current owners, procedures, contractors and competent professionals take precedence. The May 2021 floor plan is historical spatial reference only.

EIGHT CATEGORIES | TEN QUESTIONS EACH

1

bin assets, locations and waste streams

2

fill levels, collection timing and service demand

3

segregation, contamination and signage

4

hazardous, confidential and specialist waste

5

routes, lifts, loading areas and access

6

contractor coordination, storage and collections

7

weights, costs, provenance and reporting

8

prevention, reuse, recycling and emissions reduction

Scope, evidence status and non-negotiable limits

PURPOSE

An operational catalogue for waste-management teams working in Abacws. It defines the evidence and safe-response contract for decisions about assets, service demand, segregation, specialist streams, routes, contractors, reporting and prevention. It informs current owners; it does not replace local procedures or professional judgement.

PROPOSED EVIDENCE

Every sensor, integration and analytic is proposed until deployment, asset and location mapping, commissioning, permissions and current health are independently verified. An unverified, unmapped, unhealthy or unauthorised feed is unavailable evidence, not an assumed building capability.

HISTORICAL SPATIAL REFERENCE

The May 2021 floor plan may support historical orientation only. It cannot establish current room use, routes, access, services, storage, hazards, loading arrangements or closures. Current space, access, fire-safety and service owners must confirm every operational spatial fact.

AUTHORITY AND SAFE FAILURE

Current emergency direction, competent-person decisions, approved procedures, authoritative records, contracts and manifests take precedence over telemetry and analytics. The service exposes timing, provenance, permissions and conflicts, then returns supported, limited, conflicting, stale or unavailable rather than inventing an answer.

PRIVACY, CONTENTS AND SPECIALIST LIMITS

No person may be identified or blamed from waste, route, access or collection evidence. The service does not open bags, inspect confidential contents or data-bearing devices, classify unknown material, estimate exposure, or disclose restricted stores, access details, commercial terms or personal behaviour.

OPERATIONAL LIMITS

The service may retrieve facts, report observations, calculate, estimate, forecast, compare, diagnose evidence quality or recommend a bounded option. It cannot authorise handling, transport, access, spending, publication or asset release; certify compliance or safety; override current procedures; or control physical equipment.

Field guide for every question record

1

STAKEHOLDER QUESTION

A natural, distinct waste-management decision with a stable WM identifier. Each question is tied to a decision interval, a minimum evidence gate and a current accountable owner.

2

SENSORS AND TELEMETRY

Physically plausible proposed evidence stating modality and placement, spatial and temporal coverage, sampling or aggregation, calibration or functional checks, clock alignment, uptime, gaps, drift, mapping and provenance.

3

AUTHORITATIVE SUPPORTING DATA

Current source owners and precedence, with effective, observed and retrieved times, record IDs, role permissions, provenance, staleness and explicit conflict handling. Current authoritative sources always outrank historical plans and inferred state.

4

ANALYSIS REQUIRED

The operation type, decision interval, minimum evidence, method, uncertainty and conservative fallback. More data never compensates for missing authority, invalid measurement, unsafe observation or an unsupported causal claim.

5

ANSWER BOUNDARY

What the service must not infer, disclose, promise, certify, authorise, override or control. The boundary protects people, confidential contents, specialist decisions, current procedures and physical equipment.

6

WHY THIS IS USEFUL

The concrete action the waste-management team can take: inspect, prioritise, prepare, reconcile, reschedule, escalate, test or present a bounded option to the authorised owner.

OPERATION TYPES

Lookup retrieves a current source. Observation reports a verified state. Calculation applies a deterministic rule. Estimate quantifies what is not directly measured. Forecast projects a stated horizon. Comparison contrasts assets or periods. Diagnosis ranks supported explanations without claiming cause. Recommendation presents a bounded option, never authorisation.

bin assets, locations and waste streams

QUESTIONS 01-02 OF 10

WM-001

STAKEHOLDER QUESTION

Which bin and container assets are currently in service at each approved collection point, and what stream, capacity and accountable owner are confirmed for each?

WHY THIS IS USEFUL

Produces a dependable service list so the team can dispatch the correct crew, container and stream without relying on stale location assumptions.

SENSORS AND TELEMETRY

Modality and location: Paired QR or NFC scans of the asset tag and the fixed collection-point tag; controller identity only where a bin has a commissioned device. Scans occur at the asset and station, not through person-carried tracking. Coverage, timing and aggregation: Cover every approved internal station, back-of-house store and external container in the current register. Capture at installation, movement, exchange and authorised reconciliation at least every seven days; aggregate one last-confirmed state per asset and completeness totals by service zone. Measurement discipline: Require the asset and station scans in the same visit, validate tag serials against the register and treat radio silence as neither presence nor absence. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup plus observation and inventory reconciliation. Decision interval: daily for route preparation and immediately after any reported asset change. Minimum evidence: a unique asset ID, verified collection-point ID, stream, nominal capacity, current status and accountable owner. Method: join paired scans to the approved register, detect unmatched or duplicate IDs and calculate field completeness by service zone. Uncertainty: show evidence age and location confidence for each asset; mark unvisited points as uncovered rather than absent. Safe failure: return unknown for the affected asset, exclude it from automated route assumptions and request an authorised location check.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The Estates waste lead owns the asset register; Sustainability or Environmental Management owns stream definitions; Space Management owns current location records; the building manager owns local placement; the contractor owns exchange-container events. Use the approved bin register, current stream matrix, current space record and signed move or exchange records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A reported move makes the prior location stale immediately; otherwise location confirmation expires after 30 days. A signed move record or authorised physical confirmation outranks a last-seen scan; expose both states until the register owner resolves the mismatch.

ANSWER BOUNDARY

Do not treat a last-seen scan as continuous location evidence, infer current room use from the May 2021 floor plan or disclose restricted-store locations to an unauthorised role. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-002

STAKEHOLDER QUESTION

Which registered assets are missing, duplicated, moved, substituted or retired since the last confirmed reconciliation?

WHY THIS IS USEFUL

Directs a targeted asset check and prevents missed service, duplicate charging or unsafe reliance on a container that is no longer present.

SENSORS AND TELEMETRY

Modality and location: Asset-tag scans, controller heartbeats where fitted and coded movement or condition events entered by authorised staff at the collection point or store. Coverage, timing and aggregation: Cover the complete verified register, including temporary units and contractor-supplied exchange containers. Capture movement and exchange events as they occur and run an exception sweep every 24 hours; aggregate one exception state per asset with last confirmed point and evidence age. Measurement discipline: Separate silent electronics from absent assets, verify serial collisions and require a paired station scan before calling an asset moved. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: comparison followed by bounded diagnosis of record state. Decision interval: daily and before billing or route changes. Minimum evidence: one authoritative asset record plus either a valid current observation or a signed transaction event. Method: compare expected and observed identifiers, test for duplicate serials and classify only the record state as missing, possible duplicate, moved, substituted, retired or unknown. Uncertainty: state the number and age of supporting events and alternative explanations such as battery, network or scanning failure. Safe failure: leave the register unchanged and request a targeted physical check or source-owner confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Facilities owns movement and withdrawal records; Procurement owns new-asset records; the waste contractor owns exchanges; the Estates waste lead owns register changes. Use signed movement, purchase, disposal and exchange records plus current authorised observation. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Active exceptions expire after one working day; unchanged inventory evidence expires after 30 days. Do not retire or delete an asset while field and transaction evidence conflict; retain both and assign the mismatch to the register owner.

ANSWER BOUNDARY

Do not accuse cleaners, occupants or contractors of moving an asset, disclose individual access events or infer loss from a failed heartbeat alone. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

bin assets, locations and waste streams

QUESTIONS 03-04 OF 10

WM-003

STAKEHOLDER QUESTION

Does each approved waste station contain the complete current set of streams, including every documented local exception?

WHY THIS IS USEFUL

Lets the team restore a missing stream, remove an accidental extra and keep each station aligned with the authorised local service design.

SENSORS AND TELEMETRY

Modality and location: A station-tag scan, bin-tag presence and a coded external check of stream label, aperture and liner performed at the station; no bag opening or content imaging. Coverage, timing and aggregation: Cover all approved multi-stream stations in teaching, office, circulation, welfare and service areas. Capture after any station change and at least weekly; aggregate present, absent, extra and exception-coded stream sets for each station. Measurement discipline: Use the current checklist, confirm tags are attached to the observed bins and report observer disagreement rather than averaging it away. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and set comparison. Decision interval: weekly and immediately after relocation or service redesign. Minimum evidence: a verified station ID, an effective required stream set and one current external presence check. Method: compare required and observed sets, apply only valid local exceptions and list each absent, extra or mismatched element. Uncertainty: show coverage gaps, checklist age and any ambiguity where a bin tag, label and aperture disagree. Safe failure: report the station as unconfirmed and request an authorised station check before changing equipment.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability or Environmental Management owns the approved stream matrix; the Estates waste lead owns service profiles; Accessibility owns placement constraints; the local space owner owns authorised exceptions. Use the current stream standard, station register and time-bounded exception records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The stream matrix becomes stale at its effective end date; a station check expires after seven days or immediately after a move. A documented local exception outranks the generic template only for its named point and period; unresolved extra or missing streams remain exceptions.

ANSWER BOUNDARY

Do not assume colour alone establishes a stream, promise that a station serves every user need or create a local exception without owner approval. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-004

STAKEHOLDER QUESTION

Which bins are at the wrong approved collection point or assigned to the wrong internal service route?

WHY THIS IS USEFUL

Allows the team to correct route lists and placement records before crews miss a bin or service the wrong stream.

SENSORS AND TELEMETRY

Modality and location: Paired asset and collection-point scans at service time, route-stop confirmation from the crew device and fixed controller identity where commissioned; no continuous worker or trolley tracking. Coverage, timing and aggregation: Cover assets on the active route manifest and all approved points in the affected zone. Capture at each service stop and after a move; aggregate expected-versus-observed placement and route assignment by asset and route cycle. Measurement discipline: Reject unpaired or out-of-sequence scans, test route-device clock offset and distinguish a substituted container from a moved registered asset. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: comparison and recommendation. Decision interval: before each route release and after every confirmed move. Minimum evidence: a current route manifest, asset record, point record and at least one paired current observation. Method: match observed assets to expected stops, identify wrong-point and wrong-route cases and propose the smallest record or placement correction to the authorised owner. Uncertainty: state whether the discrepancy could be a temporary exchange, missed scan or stale route version. Safe failure: keep the asset on an exception list and require owner confirmation rather than redirecting crews automatically.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The Estates waste lead owns route assignments; Facilities owns current location records; the cleaning or waste service manager owns internal route manifests; the contractor owns exchange records. Use the current route plan, asset register, signed movement events and verified stop observations. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Route assignments are valid only for the stated service period; a move event makes prior placement stale. A current authorised move or exchange record outranks an old route list; where route and location owners disagree, show the mismatch and suspend automatic reassignment.

ANSWER BOUNDARY

Do not use individual access or device-location histories to attribute the move, infer current access from the historical floor plan or alter routes without the service owner. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

QUESTIONS 05-06 OF 10

Are each bin's capacity, aperture, lid, liner and handling interface compatible with its assigned stream and approved emptying method?

Do not declare a container safe for manual handling, infer material identity from the aperture or authorise a substitute liner or lifting method. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

Which bins or storage containers need urgent isolation, cleaning, maintenance or competent-person attention because of external damage, leakage, instability or obstruction?

Helps the team remove an unsafe or unusable asset from normal service and route it to the correct authorised response.

Do not open containers, diagnose chemical or biological exposure, instruct untrained handling, promise that an asset is safe or disclose restricted-store details beyond authorised roles. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

bin assets, locations and waste streams

QUESTIONS 07-08 OF 10

WM-007

STAKEHOLDER QUESTION

Are external bins, skips, compactors and waste stores in their approved locations with the required clearances and serviceable interfaces?

WHY THIS IS USEFUL

Supports safe collection preparation by identifying a misplaced container, blocked clearance or unavailable interface before a vehicle or crew arrives.

SENSORS AND TELEMETRY

Modality and location: GNSS or surveyed fixed-point location for large external assets where reception is adequate, container-tag scans, gate or door state, compactor controller status and a coded clearance check around the asset; no imaging of waste contents. Coverage, timing and aggregation: Cover all current external assets and stores in the service register. Capture location after placement or exchange, interface status at least every 15 minutes where integrated and a clearance check before each collection; aggregate one readiness state per asset. Measurement discipline: Validate GNSS against surveyed reference points, function-test discrete door and controller signals and record multipath, sheltering or communication loss explicitly. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The Estates waste lead owns approved asset positions; Space Management or the site logistics owner owns current external-use records; Security owns gate status; the contractor owns vehicle and container interfaces; Fire Safety owns required clearances. Use current site records, bookings, interface specifications and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Position evidence expires after any exchange or reported move; clearance evidence expires at the booked arrival time. The May 2021 floor plan is historical only. Current closures, fire controls and signed placement records outrank GNSS or old plans; conflicting interface states block a ready answer.

ANALYSIS REQUIRED

Operation: lookup, observation and readiness calculation. Decision interval: before each external collection and after any placement change. Minimum evidence: a current approved location, asset identity, required clearance, interface specification and current access or obstruction state. Method: compare observed position and clearance against current authorised tolerances and evaluate each required interface separately. Uncertainty: state location accuracy, unobserved sides, status age and any dependency on unauthorised or unhealthy integrations. Safe failure: return not confirmed, hold the collection preparation and request the responsible site owner to verify the condition.

ANSWER BOUNDARY

Do not infer current yard use from the historical plan, declare a vehicle manoeuvre safe, expose gate-control details broadly or activate a compactor, door or gate. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-008

STAKEHOLDER QUESTION

Which collection points show a persistent service gap or excess capacity large enough to justify a formal proposal to add, remove, resize or relocate a station?

WHY THIS IS USEFUL

Lets the team prepare a focused station-change proposal with evidence of demand, service burden, accessibility and contamination risk.

SENSORS AND TELEMETRY

Modality and location: Commissioned fill or weight sensing where suitable, collection-reset scans, coded overflow and contamination events, and privacy-preserving zone-level footfall counts where already authorised; location checks occur at the station. Coverage, timing and aggregation: Cover candidate stations and at least one comparable point over a minimum of six representative service cycles, including relevant teaching or event periods. Aggregate fill-at-service, overflow episodes, service stops, contamination rate and zone demand by day and stream. Measurement discipline: Validate fill geometry after liner or bin changes, test reset detection against signed service events and treat zone counts only as contextual totals, not as a proxy for individual behaviour. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The Estates waste lead owns station configuration; Sustainability owns stream policy; Accessibility owns access constraints; Space Management and the local owner confirm current use; the service manager owns collection records. Use current asset, route, incident, access and service-demand records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Evidence must cover the current operating pattern and becomes stale after a material room-use, route or timetable change. Current accessibility or fire constraints override an apparent demand benefit. Conflicting demand measures remain separate and reduce confidence rather than being averaged without basis.

ANALYSIS REQUIRED

Operation: comparison, estimate and bounded recommendation. Decision interval: monthly, or before a planned space or service change. Minimum evidence: six representative cycles, valid station mapping, current service records and a current location and accessibility check. Method: compare demand and service burden with comparable stations, test practical alternatives and present a change option with expected range rather than a single asserted outcome. Uncertainty: show seasonality, event effects, missing cycles and the possibility that observed problems arise from servicing or signage rather than capacity. Safe failure: state that evidence is insufficient and continue the current authorised arrangement while collecting targeted data.

ANSWER BOUNDARY

Do not use aggregate footfall to infer who generated waste, relocate equipment automatically, promise a reduction in contamination or override accessibility, fire or local-space decisions. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

bin assets, locations and waste streams

QUESTIONS 09-10 OF 10

WM-009

STAKEHOLDER QUESTION

Where do stream names, colours, icons, apertures, liners or system records conflict within a station or between neighbouring stations?

WHY THIS IS USEFUL

Enables a precise correction list so users receive consistent cues and crews do not service visually similar containers as different streams.

SENSORS AND TELEMETRY

Modality and location: A station-tag scan and coded external comparison of the displayed label, colour reference, icon code, aperture, liner and bin tag; spectral or colour measurement only where a controlled reference card and lighting condition are used. Coverage, timing and aggregation: Cover all approved stations after sign or bin changes and at least every 30 days, with neighbouring stations compared within the same visit. Aggregate each cue as current, conflicting, missing or unknown for the station. Measurement discipline: Check colour instruments against a reference card, record ambient lighting for any colour result and use exact code values rather than free-text descriptions. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and attribute-by-attribute comparison. Decision interval: monthly and after any signage, liner or container change. Minimum evidence: a verified station ID, current approved cue set and one complete external observation. Method: compare exact codes for name, colour, icon, aperture, liner and backend stream; produce a field-level correction list. Uncertainty: show whether colour results are instrumented or visual, the lighting condition and any ambiguous or inaccessible cue. Safe failure: mark the station inconsistent and request the stream-standard owner to issue the authorised correction.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability owns the stream vocabulary and visual standard; the Estates waste lead owns station configuration; Communications or Brand owners control approved artwork; the contractor owns liner and collection specifications. Use current artwork files, stream matrix, asset register and approved local exceptions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Displayed-cue evidence expires after any replacement and otherwise after 30 days. The current approved artwork and stream matrix outrank legacy labels. If label, bin tag and contractor stream code disagree, expose every value and block an automatic correction until owners agree.

ANSWER BOUNDARY

Do not infer a stream from colour alone, inspect deposited material, disclose draft artwork or change signs, liners or system mappings without owner authority. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-010

STAKEHOLDER QUESTION

Which temporary or event waste assets are active now, what service period and stream apply, and who is accountable for removal and final reconciliation?

WHY THIS IS USEFUL

Prevents temporary bins from being missed, left behind or absorbed into the permanent register without an authorised owner.

SENSORS AND TELEMETRY

Modality and location: Temporary asset and station tags, event-deployment scans, optional fill sensing only on commissioned units and coded setup and removal confirmations at the approved temporary location. Coverage, timing and aggregation: Cover every temporary asset listed for the event or works period. Capture at issue, placement, each service visit, relocation and removal; aggregate current state, stream, service window, point and responsible owner per asset. Measurement discipline: Use temporary identifiers that cannot collide with permanent assets, verify the planned end date and treat a missing removal scan as unresolved rather than proof that the asset remains on site. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup, observation and lifecycle reconciliation. Decision interval: at setup, each service cycle, event close and final hand-back. Minimum evidence: an authorised temporary-asset record, current location, stream, active period and accountable owner. Method: join deployment, service and removal events into a lifecycle state and identify unplaced, unserviced, relocated or unreturned assets. Uncertainty: state missing event scans, uncertain location and any period where sensing was not commissioned or healthy. Safe failure: leave the asset on the exception list, prevent automatic transfer to the permanent register and request owner confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The event or project owner owns the need and service period; the Estates waste lead owns deployment approval; the service manager owns collection arrangements; Facilities owns current location; the contractor owns hired units. Use the authorised event plan, temporary asset register, service booking and signed hand-back record. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Temporary records expire at the approved end time unless extended by the owner. A current closure or safety direction overrides the event plan. If the event list and observed assets differ, keep the exceptions visible until the event and waste owners reconcile them.

ANSWER BOUNDARY

Do not infer event attendance or individual behaviour, use the May 2021 plan as current placement authority, extend the service period or dispose of an asset without the accountable owners. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

fill levels, collection timing and service demand

QUESTIONS 01-02 OF 10

WM-011

STAKEHOLDER QUESTION

Which bins should be emptied on the next internal service route, and which can safely remain until the following confirmed visit?

WHY THIS IS USEFUL

Creates a bounded stop list that reduces unnecessary handling while protecting against overflow before the next authorised service.

SENSORS AND TELEMETRY

Modality and location: Under-lid time-of-flight or ultrasonic fill sensing where bin geometry supports it, load cells on suitable fixed containers, lid-state switches and a paired service-reset scan at the bin and route stop. Coverage, timing and aggregation: Cover only commissioned assets on the next route. Sample fill at one to five minute intervals for rapidly used stations or every 15 minutes for stable points; aggregate the latest valid fill, recent fill rate, reset state and threshold margin per bin. Measurement discipline: Revalidate geometry after liner, lid or bin changes, compare fill readings with periodic reference checks and reject values affected by tilted contents, sensor blockage or a missing emptying reset. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: observation, short-horizon calculation and recommendation. Decision interval: before each route release and whenever a high-priority alert arrives. Minimum evidence: current route and next-visit time, one valid recent fill measure or authorised observation, asset capacity and approved threshold. Method: calculate threshold margin to the next visit, apply stream and location service rules and order stops by risk without changing the authorised route boundary. Uncertainty: show measurement error, fill-shape sensitivity, forecast horizon and uncovered bins; do not present a single exact time-to-full where evidence is weak. Safe failure: retain the scheduled stop or request a physical check; never recommend skipping a bin with stale, missing or conflicting evidence.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The service manager owns route rules and emptying thresholds; the Estates waste lead owns asset mappings; the building manager owns local exceptions; cleaning records provide signed service events. Use current route manifests, thresholds, asset data, closures and service confirmations. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Latest fill must be within the stream-specific freshness window, normally 30 minutes for high-demand points and two hours for stable points. A current hygiene, safety or local instruction overrides fill optimisation. Where sensing and a verified overflow report conflict, the report takes precedence and both are retained.

ANSWER BOUNDARY

Do not promise that a bin will not overflow, infer who used it, instruct staff to compact material by hand or dispatch, cancel or control a service without the authorised manager. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-012

STAKEHOLDER QUESTION

Which collection points are likely to cross their approved fill threshold before the next confirmed internal or contractor collection?

WHY THIS IS USEFUL

Lets the team add capacity, bring forward a visit or warn the authorised owner before an avoidable overflow develops.

SENSORS AND TELEMETRY

Modality and location: Commissioned fill or weight sensing at the container, verified emptying resets and approved service-event timestamps; event or timetable context may be used only as aggregate operational context. Coverage, timing and aggregation: Cover each candidate point continuously or at an interval fine enough to detect its normal rise, with at least three complete comparable fill cycles. Aggregate recent rate, cycle variability and projected threshold-crossing window for each stream and point. Measurement discipline: Exclude cycles with unconfirmed resets, sensor obstruction, mapping changes or long gaps; compare sensor output with reference observations at commissioning and after material geometry changes. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: forecast with scenario comparison. Decision interval: hourly for high-demand points and at least daily for other active points. Minimum evidence: three comparable complete cycles, current capacity and threshold, a confirmed next collection window and no unresolved sensor-health fault. Method: fit a robust local rate or quantile model, produce earliest, central and latest crossing windows and test one plausible surge scenario where justified. Uncertainty: show interval width, cycle count, excluded periods and sensitivity to the next collection time. Safe failure: return forecast unavailable and retain the normal schedule while requesting a current physical check or manual estimate.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The service manager owns collection times and threshold policy; the contractor owns confirmed external collection windows; the Estates waste lead owns capacity and mapping; event or timetable owners supply current context. Use signed bookings, service events, current thresholds and verified sensor health records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A forecast is valid only until the earlier of the next material event, collection, route change or stated horizon. Confirmed collection changes outrank prior schedules. If event context and measured fill diverge, measured evidence remains primary and the divergence is shown.

ANSWER BOUNDARY

Do not state an exact overflow time as certain, treat calendar events as proof of waste generation or alter contractor bookings without authorised confirmation. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

fill levels, collection timing and service demand

QUESTIONS 03-04 OF 10

WM-013

STAKEHOLDER QUESTION

Where are overfill, lid-open, blockage, leakage or delayed-emptying conditions recurring rather than isolated?

WHY THIS IS USEFUL

Identifies locations that need a sustained service, asset or placement response rather than repeated one-off callouts.

SENSORS AND TELEMETRY

Modality and location: Fill, lid, tilt or leak signals where suitable, paired service-reset scans and coded external incident observations at the bin or store; no imaging of deposited material. Coverage, timing and aggregation: Cover active points with at least eight service cycles or eight weeks of evidence. Capture signals at their commissioned rate and incidents when observed; aggregate episodes by condition, duration, point, stream and service cycle. Measurement discipline: Debounce short lid events, validate reset times against signed service events and separate device faults from physical episodes using health and reference checks. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The service manager owns service exceptions; Facilities owns asset defects; the Estates waste lead owns station configuration; the building manager owns local incident records. Use current service logs, defect tickets, verified observations and commissioned telemetry. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Episodes remain open until a current service or condition confirmation closes them. Changes in capacity, route or sensor mapping split the comparison period. If incident records and telemetry disagree, show both and classify the episode as conflicting rather than selecting one silently.

ANALYSIS REQUIRED

Operation: comparison and bounded diagnosis. Decision interval: weekly, with immediate escalation for a current serious external condition. Minimum evidence: at least three supported episodes of the same condition or a stated policy trigger, plus current asset and service mappings. Method: group validated episodes, compare frequency and duration with comparable points and rank supported explanations such as service timing, capacity, asset fault or access delay without asserting cause. Uncertainty: show episode definition, missing cycles, exposure time and changes in service conditions. Safe failure: report insufficient evidence for recurrence and continue event-level handling while collecting targeted observations.

ANSWER BOUNDARY

Do not attribute recurring conditions to named occupants or crews, infer bag contents, diagnose exposure or declare a root cause without evidence that distinguishes competing explanations. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-014

STAKEHOLDER QUESTION

Which fill, lid, weight or emptying-reset signals are unreliable enough to require a physical device check before operational use?

WHY THIS IS USEFUL

Stops weak telemetry from driving missed collections, false overflow alerts or misleading performance figures.

SENSORS AND TELEMETRY

Modality and location: Device self-test, battery and communication status, reference-distance or test-weight checks, paired reset scans and gateway receipt records at each commissioned sensor and controller. Coverage, timing and aggregation: Cover every device used for operational decisions. Collect health data at least every 15 minutes, run a documented reference check at commissioning and on the device-specific schedule, and aggregate availability, gap length, offset, drift and implausible-state counts per device. Measurement discipline: Use known empty and reference states, check clock offset and detect stuck values, impossible jumps, excessive noise, missed resets and drift beyond the approved tolerance. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The sensor service owner owns commissioning and health records; the Estates waste lead owns asset mapping; the service manager owns operational acceptance thresholds; suppliers own device specifications. Use current commissioning certificates, check records, mapping tables and incident tickets. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Health state expires at the next failed self-test, mapping change or tolerance breach. A verified physical observation outranks an unhealthy signal. If supplier limits and local acceptance limits differ, apply the more conservative current authorised limit and expose the conflict.

ANALYSIS REQUIRED

Operation: diagnosis of evidence quality and recommendation. Decision interval: continuous health screening with a daily exception list. Minimum evidence: device identity and mapping, current health telemetry, at least one valid reference check and the approved tolerance. Method: score each device for availability, calibration or functional status, clock offset, gaps and drift; classify it as usable, restricted, suspect or unavailable for the named decision. Uncertainty: show the failed tests, data period, tolerance source and whether the fault affects fill, timing, reset or all derived measures. Safe failure: remove the signal from calculations, retain the scheduled service rule and request an authorised device check.

ANSWER BOUNDARY

Do not remotely recalibrate or reset equipment, infer a physical bin condition from a device fault or suppress a verified external safety report because telemetry appears normal. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

fill levels, collection timing and service demand

QUESTIONS 05-06 OF 10

WM-015

STAKEHOLDER QUESTION

What is the normal fill and collection-reset pattern for each stream, location and comparable operating period?

WHY THIS IS USEFUL

Provides a defensible baseline for staffing, route timing, anomaly screening and later service-change proposals.

SENSORS AND TELEMETRY

Modality and location: Commissioned fill or weight series, verified service-reset scans and approved aggregate context such as teaching weeks, closure periods and confirmed events. Coverage, timing and aggregation: Cover each active stream and point for at least eight complete representative cycles, preferably across multiple weeks. Aggregate median and percentile fill trajectories, cycle length, fill at service and reset reliability by point, stream and comparable period. Measurement discipline: Exclude unconfirmed emptying events and unhealthy device periods, split data after capacity or mapping changes and test reference states at the start and end of the baseline window. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: calculation and comparison. Decision interval: monthly, and whenever service conditions materially change. Minimum evidence: eight complete valid cycles, stable mapping and capacity, confirmed resets and an effective operating-period label. Method: calculate robust central and percentile trajectories, variability and fill-at-service; compare only points and periods with matched definitions. Uncertainty: show sample size, excluded cycles, seasonal coverage and the range of normal rather than a single average. Safe failure: state that no stable baseline exists and use current fixed service rules until sufficient comparable evidence is available.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The service manager owns service-event records; the Estates waste lead owns capacity and mappings; timetable, event and building owners supply effective operating-period labels. Use current asset records, route logs, closure records and verified telemetry. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A baseline is valid only for its stated capacity, route, stream and operating context. A material service or space-use change ends the prior baseline. Conflicting period labels remain explicit and prevent automatic pooling.

ANSWER BOUNDARY

Do not interpret aggregate demand as individual behaviour, combine unlike periods for convenience or present a baseline as a guaranteed future pattern. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-016

STAKEHOLDER QUESTION

Which current service frequencies are consistently too early or too late relative to the approved fill and hygiene target band?

WHY THIS IS USEFUL

Supports a focused schedule adjustment proposal while preserving mandatory service, hygiene and local constraints.

SENSORS AND TELEMETRY

Modality and location: Fill or weight evidence at the bin, verified emptying resets, external hygiene-condition codes and route-stop timestamps; no content classification or person-level monitoring. Coverage, timing and aggregation: Cover at least six valid service cycles for each candidate point and include all mandatory visits. Aggregate fill-at-service, time above the upper target, time below the lower target and hygiene exceptions by point and stream. Measurement discipline: Validate target-band thresholds against current policy, exclude faulty or gapped cycles and confirm that a low reading is not caused by a failed reset or moved sensor. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: comparison and bounded recommendation. Decision interval: monthly or before the next schedule cycle. Minimum evidence: six valid cycles, current target band, confirmed service events and no unresolved hygiene or sensor-health exception. Method: compare fill-at-service and threshold exposure with the approved band, separate early, balanced and late service patterns and present a schedule option for owner decision. Uncertainty: show cycle variability, mandatory constraints, missed evidence and the expected range of change rather than an asserted saving. Safe failure: retain the current frequency and request additional cycles or an authorised physical condition check.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The service manager owns service frequency and target bands; Hygiene or Facilities owners set mandatory visit rules; the Estates waste lead owns capacity and mapping; the contractor owns external collection constraints. Use current schedules, procedures, service events and verified evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Findings expire after a capacity, use, route or hygiene-rule change. Mandatory service rules outrank apparent efficiency. Where fill evidence supports less frequent service but hygiene records do not, the more conservative requirement prevails and the conflict is shown.

ANSWER BOUNDARY

Do not cancel mandatory visits, promise cost savings, treat low fill as proof that a point is unnecessary or alter staff rotas or contractor frequencies without authority. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

fill levels, collection timing and service demand

QUESTIONS 07-08 OF 10

WM-017

STAKEHOLDER QUESTION

What temporary capacity and service pattern is justified for a confirmed event, teaching surge, project move or closure?

WHY THIS IS USEFUL

Helps the team prepare enough temporary capacity and collection support without installing unnecessary assets or assuming attendance.

SENSORS AND TELEMETRY

Modality and location: Temporary-bin tags, fill or weight sensing only on commissioned units, service-reset scans and approved aggregate event or timetable parameters; measurements remain at the container or zone level. Coverage, timing and aggregation: Cover the affected points during setup, the event window and run-down, plus at least one relevant historic comparator where definitions match. Aggregate expected demand range, current fill, temporary capacity and required service checkpoints by stream and zone. Measurement discipline: Verify temporary sensor mapping at deployment, use reference checks after movement and do not translate attendance estimates directly into waste without an evidenced conversion range. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The event or project owner owns the confirmed scope and period; the Estates waste lead owns temporary capacity approval; the service manager owns staffing and routes; Facilities owns current closures and access. Use authorised event plans, current route capacity, comparable service records and live asset status. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The plan expires at the event end or on a scope, access or attendance change. Current safety or access directions override the planned layout. If comparable history is weak, keep the demand range wide and identify it as an estimate.

ANALYSIS REQUIRED

Operation: estimate, scenario calculation and recommendation. Decision interval: during event planning, at setup and at agreed checkpoints during the event. Minimum evidence: a confirmed scope and time window, current affected assets, at least one defensible demand basis and available service capacity. Method: build low, central and high demand scenarios, test temporary capacity and service checkpoints against thresholds and list operational dependencies. Uncertainty: show conversion assumptions, comparator differences, event uncertainty and uncovered streams or locations. Safe failure: use the conservative current service plan, request manual checks at defined checkpoints and avoid a precise demand claim.

ANSWER BOUNDARY

Do not infer individual attendance or behaviour, guarantee adequate capacity, place temporary assets in unconfirmed locations or authorise extra labour, transport or spending. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-018

STAKEHOLDER QUESTION

How should today's service priorities change after a confirmed missed, late or partial collection?

WHY THIS IS USEFUL

Gives the team an ordered recovery list based on current capacity, hygiene and access risk rather than first-reported order alone.

SENSORS AND TELEMETRY

Modality and location: Current fill or weight evidence, service-reset state, coded external hygiene or leakage observations and confirmed collection-event status at affected bins, stores and bulk containers. Coverage, timing and aggregation: Cover every point dependent on the missed or partial service. Refresh high-risk points at least every 15 minutes where sensing exists and other points through authorised checks; aggregate remaining capacity, threshold margin, current condition and next feasible service window. Measurement discipline: Confirm that the collection failure is authoritative, validate fill evidence after any partial emptying and separate a stale reset from a genuinely unemptied container. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The service manager owns recovery priorities; the contractor owns external service status; Facilities and the building manager own access and local condition; Health and Safety owns urgent controls. Use signed missed-service records, current capacity, access status and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The recovery ranking expires when a point is serviced, access changes or evidence exceeds its freshness limit. Emergency and hygiene directions outrank optimisation. Conflicting contractor and site records remain visible and trigger source-owner confirmation.

ANALYSIS REQUIRED

Operation: observation, calculation and recommendation. Decision interval: immediately after confirmation and at each recovery checkpoint. Minimum evidence: a confirmed service exception, current status for each affected point, capacity and threshold, and a feasible service or storage option. Method: calculate urgency from remaining capacity, condition, stream, time to next feasible service and consequences of delay; provide an ordered list with reasons. Uncertainty: show stale or uncovered points, uncertain partial-emptying status and sensitivity to the expected recovery time. Safe failure: apply the standing contingency plan, physically check unobserved high-risk points and escalate to the authorised owner.

ANSWER BOUNDARY

Do not dispatch or cancel crews or contractors, instruct unsafe manual handling, promise a recovery time or disclose restricted access arrangements beyond authorised roles. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

fill levels, collection timing and service demand

QUESTIONS 09-10 OF 10

WM-019

STAKEHOLDER QUESTION

Is any skip, compactor or bulk container approaching an approved operating or collection threshold that needs competent action?

WHY THIS IS USEFUL

Provides early warning for a bulk asset while keeping operating decisions with the trained owner and manufacturer-approved controls.

SENSORS AND TELEMETRY

Modality and location: Manufacturer-supported load cells, hydraulic pressure or compaction-cycle counts, door or lock state and external level sensing where suitable, all at the bulk asset controller or structure. Coverage, timing and aggregation: Cover only assets with a verified interface and current commissioning. Sample at the manufacturer-supported rate and aggregate current load or level, cycle count, alarm state, threshold margin and data-health state per asset. Measurement discipline: Use traceable test loads or manufacturer functional checks, validate cycle-to-load relationships locally and reject pressure or cycle counts as a direct mass measure unless the approved conversion is evidenced. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The competent plant owner owns operating thresholds and impairment status; the Estates waste lead owns collection thresholds; the contractor owns exchange criteria; the manufacturer supplies limits. Use current operating instructions, maintenance status, bookings and controller records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Threshold evidence expires at the next compaction cycle, door event or controller fault. Competent-person direction and manufacturer alarms outrank analytics. If different measures disagree, present the most conservative state and block a normal-status answer.

ANALYSIS REQUIRED

Operation: observation and threshold calculation. Decision interval: continuous where integrated and before every planned collection or exchange. Minimum evidence: verified asset identity, current health, approved operating and collection thresholds and at least one valid current measure. Method: calculate margin to each threshold separately and distinguish operating limit, collection trigger, alarm and unknown state. Uncertainty: show measurement tolerance, conversion uncertainty, last valid sample and any dependence on estimated density. Safe failure: return status unknown, keep the asset out of automated planning and refer to the competent plant owner.

ANSWER BOUNDARY

Do not start, stop, reset or compact equipment, tell anyone to enter or open the asset, convert cycles to mass without evidence or declare the plant safe or compliant. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-020

STAKEHOLDER QUESTION

Which service points show a sustained demand change large enough to support a capacity or collection-frequency proposal?

WHY THIS IS USEFUL

Separates lasting demand shifts from short events so the team can target a formal service change at the right points.

SENSORS AND TELEMETRY

Modality and location: Commissioned fill or weight history, verified service resets, overflow and hygiene events, asset-capacity records and approved aggregate operating-context labels. Coverage, timing and aggregation: Cover at least 12 representative service cycles before and after the suspected change, with stable mapping or an explicit adjustment for changes. Aggregate fill generation per cycle, fill-at-service, threshold episodes and service effort by point and stream. Measurement discipline: Split data at sensor, capacity, route or stream changes, check reference states across the period and avoid treating a short event as a sustained shift. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The service manager owns frequency decisions; the Estates waste lead owns capacity and station configuration; Sustainability owns stream policy; building and timetable owners supply current context. Use current route logs, asset records, incident records and verified telemetry. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A change result is valid only for the defined operating context and expires after another material change. Current hygiene, access or accessibility constraints override a capacity-based option. Conflicting trend measures remain separate and reduce the recommendation strength.

ANALYSIS REQUIRED

Operation: comparison, change detection and recommendation. Decision interval: monthly or at the end of a representative operating block. Minimum evidence: 12 valid cycles on each side where possible, stable definitions, confirmed resets and current constraints. Method: compare robust per-cycle demand and threshold exposure, test a sustained level or slope change and model the effect of feasible capacity or frequency options. Uncertainty: show confidence interval, missing periods, sensitivity to event exclusions and whether evidence supports direction only or also magnitude. Safe failure: state that the change is not established and keep the current authorised service while collecting further representative cycles.

ANSWER BOUNDARY

Do not claim a causal reason for the shift, infer occupancy at person level, promise savings or change capacity, routes or contracts without authorised decisions. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

segregation, contamination and signage

QUESTIONS 01-02 OF 10

WM-021

STAKEHOLDER QUESTION

Which stations or collected loads have enough current evidence to confirm an ordinary-stream contamination problem that needs action?

WHY THIS IS USEFUL

Prevents the team from acting on weak alerts while allowing a supported contamination issue to be contained, relabelled or escalated promptly.

SENSORS AND TELEMETRY

Modality and location: Coded observations by authorised trained staff at the station or controlled transfer point, contractor rejection or contamination codes, load weights and optional non-imaging aperture or lid events. Observations must not open bags or record readable discarded material. Coverage, timing and aggregation: Cover the named station or traceable load and the relevant service event. Capture observations at collection or transfer and contractor outcomes when received; aggregate confirmed, possible, rejected and unresolved contamination states by stream and event. Measurement discipline: Use a versioned ordinary-contamination code set, require observer training and record whether only the visible external or top surface was accessible; do not use a sensor signal as material identity. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability or Environmental Management owns ordinary-stream acceptance rules; the service manager owns station observations; the contractor owns rejection and facility codes; the Estates waste lead owns load-to-station mappings. Use current acceptance rules, service records and traceable contractor evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A station observation expires at the next emptying; a load finding applies only to that traceable load. A contractor rejection code outranks a local estimate for the load, while a current approved acceptance rule outranks legacy contractor wording. Broken load linkage prevents attribution to a station.

ANALYSIS REQUIRED

Operation: observation and evidence-sufficiency decision. Decision interval: at each reported alert, collection or contractor rejection. Minimum evidence: a current stream rule, traceable station or load, at least one authorised observation or contractor outcome and valid event timing. Method: apply an evidence gate that separates confirmed ordinary contamination, possible contamination, non-actionable alert and unknown; list the supporting facts. Uncertainty: show whether evidence is station-level or load-level, the observable surface, linkage quality and any conflicting codes. Safe failure: keep the alert unconfirmed, avoid corrective claims and request an authorised external check or contractor clarification.

ANSWER BOUNDARY

Do not open bags, read or image discarded documents, identify a source person or group, classify unknown or hazardous material or treat a load finding as proof about every contributing station. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-022

STAKEHOLDER QUESTION

What ordinary contamination category is actually supported for each actionable alert, and which alerts remain too uncertain to classify?

WHY THIS IS USEFUL

Allows the team to choose the correct ordinary-stream response instead of applying a generic contamination label.

SENSORS AND TELEMETRY

Modality and location: A versioned coded observation of visible ordinary material classes by authorised staff, contractor material or rejection codes and stream-specific weight or moisture evidence where explicitly accepted by the owner; no automated image recognition of waste. Coverage, timing and aggregation: Cover only alerts that passed the preceding evidence-sufficiency gate. Capture one code set per observation or contractor event and aggregate the supported category, competing category and unclassified state for the named station or load. Measurement discipline: Use mutually defined categories, record observer confidence and observable limits, and treat moisture, weight or aperture signals as supporting context rather than material classification. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability owns the ordinary-contamination taxonomy; the contractor owns facility codes; the service manager owns local observation records. Use the current stream acceptance matrix, code translation table and traceable event evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Category support expires with the event and cannot be carried to future loads. If local and contractor code sets map differently, show both original codes and the approved translation; where no approved mapping exists, return unclassified.

ANALYSIS REQUIRED

Operation: classification of record state using controlled ordinary categories. Decision interval: for each actionable contamination event. Minimum evidence: a traceable event, current taxonomy, one supported observation or contractor code and no indication of unknown specialist material. Method: map only observed or reported ordinary categories to the approved code set, preserve alternative categories and refuse classification beyond the evidence. Uncertainty: state observer confidence, visible scope, code-mapping quality and whether the category is direct or translated. Safe failure: return ordinary contamination - unclassified and refer the event to the authorised owner without suggesting handling.

ANSWER BOUNDARY

Do not identify specific discarded items where that could reveal confidential information, infer intent or blame, or classify chemicals, sharps, batteries, biological material or any unknown substance. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

segregation, contamination and signage

QUESTIONS 03-04 OF 10

WM-023

STAKEHOLDER QUESTION

Which label, colour, icon, aperture, liner or station pairing conflicts with the current segregation rules?

WHY THIS IS USEFUL

Gives the team an exact physical and records correction list before inconsistent cues create further sorting errors.

SENSORS AND TELEMETRY

Modality and location: Station-tag scan and coded external inspection of each cue, with controlled colour measurement only where a reference card and lighting record are used; system mappings are read from the authorised configuration source. Coverage, timing and aggregation: Cover every stream element at the named station and its paired neighbour. Capture after any change and at least every 30 days; aggregate each cue as matched, mismatched, missing, obscured or unknown. Measurement discipline: Verify the checklist version, use exact artwork or code identifiers, record ambient light for colour measurements and never infer the stream from the deposited material. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and field-level comparison. Decision interval: monthly and immediately after a signage, liner, aperture or stream change. Minimum evidence: verified station ID, current approved cue set and a complete external observation of the named elements. Method: compare exact names, codes and physical attributes, then issue a correction list ordered by potential service impact. Uncertainty: state unobserved faces, lighting or access limits and any cue that cannot be matched to a current approved source. Safe failure: mark the station unconfirmed for segregation messaging and request the authorised owner to specify the correction.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability owns segregation rules and cue standards; Communications owns approved artwork; the Estates waste lead owns station configuration; the contractor owns liner and collection code requirements. Use current artwork, stream matrix, asset and system configuration records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A cue check expires immediately after replacement and otherwise after 30 days. Current approved standards outrank legacy labels. If physical and digital mappings differ, retain both, mark the station inconsistent and refer the mapping decision to the named owners.

ANSWER BOUNDARY

Do not change labels or system mappings automatically, infer a stream from colour alone, inspect contents or disclose unpublished artwork or restricted configuration data. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-024

STAKEHOLDER QUESTION

Which waste signs are missing, obscured, damaged, outdated, poorly positioned or unreadable from the normal approach?

WHY THIS IS USEFUL

Supports a targeted sign replacement and placement list rather than blanket replacement across the building.

SENSORS AND TELEMETRY

Modality and location: A station-tagged external sign check, mounting-height and viewing-distance measurement, ambient illuminance reading and optional contrast measurement against the approved artwork; no camera capture of nearby people or waste. Coverage, timing and aggregation: Cover all approved station approaches and the positions from which a user normally chooses a stream. Capture after placement changes and every 60 days; aggregate sign presence, condition, version, height, illumination and obstruction state per approach. Measurement discipline: Use a calibrated tape or laser distance tool, traceable illuminance meter and controlled contrast method; record temporary obstructions separately from the fixed sign state. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: observation, lookup and recommendation. Decision interval: every 60 days and after any reported sign or route change. Minimum evidence: station and approach identity, current approved sign version and one complete current external measurement set. Method: compare presence, version, condition, mounting and visibility attributes with the current criteria and rank actions as replace, uncover, reposition for consideration or verify. Uncertainty: show measurement tolerance, lighting range, temporary obstructions and approaches that could not be assessed. Safe failure: return sign status unconfirmed and request a current authorised check before ordering or relocating material.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Communications owns approved sign content and versions; Sustainability owns waste instructions; Accessibility owns inclusive-information criteria; the local space owner owns placement constraints. Use current artwork, sign register, accessibility criteria and authorised location records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Condition evidence expires after a reported change or after 60 days. The current approved artwork version outranks the displayed legacy sign, but the local owner must confirm placement. A temporary obstruction does not justify permanent relocation without current repeated evidence.

ANSWER BOUNDARY

Do not record identifiable passers-by, infer comprehension from physical measurements, promise accessibility for every user or move signs without the space and accessibility owners. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

segregation, contamination and signage

QUESTIONS 05-06 OF 10

WM-025

STAKEHOLDER QUESTION

Which station layouts, neighbouring streams or approach arrangements are repeatedly associated with segregation errors and merit a controlled local test?

WHY THIS IS USEFUL

Helps the team target one high-value physical arrangement test without claiming that layout alone caused the errors.

SENSORS AND TELEMETRY

Modality and location: Station configuration records, coded ordinary-contamination events, stream-specific collection weights, service resets and privacy-preserving zone-level flow counts where already authorised; no person or journey tracking. Coverage, timing and aggregation: Cover at least six comparable service cycles for candidate and comparison stations, with stable stream definitions. Aggregate error events, contamination rate, service demand and arrangement attributes by station and cycle. Measurement discipline: Confirm load-to-station linkage, split data after configuration changes and treat aggregate flow only as exposure context; do not derive individual behaviour or material identity from sensors. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The Estates waste lead owns station layout; Sustainability owns segregation rules; Accessibility and Fire Safety own placement constraints; the service manager and contractor own event evidence. Use current station records, contamination events, service logs and approved flow totals. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Associations expire after a material layout, stream, route or use change. Accessibility and fire constraints outrank a statistical association. If station-level and load-level evidence cannot be linked reliably, exclude the affected cycles rather than assigning them.

ANALYSIS REQUIRED

Operation: comparison and bounded diagnosis leading to a test recommendation. Decision interval: monthly or before a planned local arrangement test. Minimum evidence: six comparable cycles, stable definitions, traceable event linkage and a feasible comparison station or before-period. Method: compare error rates across defined arrangement attributes, adjust only for documented context and recommend one testable change with a pre-stated measure. Uncertainty: show sample size, linkage loss, confounding service differences and that association does not establish cause. Safe failure: state that no layout-specific test is supported and focus on data quality or general signage checks instead.

ANSWER BOUNDARY

Do not identify or profile users, claim a behavioural cause, place equipment in an unconfirmed route or access space, or implement a test without the current owners. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-026

STAKEHOLDER QUESTION

Which station signs and physical cues need an accessibility assessment or an authorised alternative format before replacement or relocation?

WHY THIS IS USEFUL

Ensures that waste messaging changes are sent to the right accessibility owner with concrete physical evidence and no unsupported claim about user capability.

SENSORS AND TELEMETRY

Modality and location: External measurements of mounting height, viewing distance, text size, luminance contrast and ambient light, plus coded presence of tactile, symbol or alternative-format provisions where specified; no facial, gaze or behaviour sensing. Coverage, timing and aggregation: Cover all approaches and interaction sides defined by the current station record. Capture after a sign or station change and on the accessibility owner's schedule; aggregate measured attributes and unmet or unknown criteria per approach. Measurement discipline: Use calibrated measurement tools and current test methods, record reach or view obstructions and distinguish a missing measurement from a failed criterion. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Accessibility owns inclusive-information and reach criteria; Communications owns artwork; Sustainability owns waste instructions; Space Management and the local owner confirm current approach geometry. Use current criteria, artwork and current measured layout records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Measurements become stale after relocation, furniture change or lighting change. Current competent accessibility direction outranks generic criteria. The May 2021 floor plan cannot establish current approaches. Conflicting criteria are shown and sent to the accessibility owner.

ANALYSIS REQUIRED

Operation: lookup, measurement comparison and recommendation. Decision interval: at design selection, installation and after any material approach change. Minimum evidence: current sign version, current station geometry, complete measurements for each relevant approach and the effective criteria. Method: compare each measured attribute with the named criterion and identify whether the next action is redesign, alternative format, relocation consideration or further assessment. Uncertainty: show measurement tolerance, inaccessible approaches and criteria that require user or professional judgement beyond sensing. Safe failure: return not assessed and refer the design to the accessibility owner before procurement or placement.

ANSWER BOUNDARY

Do not infer disability, comprehension, visual attention or preference, promise universal usability or authorise a sign, format or station location. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

segregation, contamination and signage

QUESTIONS 07-08 OF 10

WM-027

STAKEHOLDER QUESTION

Where are food residues, liquids or wet conditions repeatedly compromising recoverable dry recycling?

WHY THIS IS USEFUL

Lets the team target lids, drainage, station pairing, service timing or user guidance where dry recycling is most at risk.

SENSORS AND TELEMETRY

Modality and location: External leak or spill sensors near the station base where suitable, bin headspace humidity or moisture indicators only if validated for the container, coded authorised observations and contractor wet-load or rejection codes; no bag opening or content imaging. Coverage, timing and aggregation: Cover dry-recycling stations with reported wetting and their traceable loads for at least six service cycles. Aggregate wet-condition events, external leaks, contractor outcomes and service timing by station and cycle. Measurement discipline: Validate moisture indicators against controlled reference conditions, account for ambient humidity and cleaning events and never convert a humidity signal directly into a contamination category. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability owns dry-recycling acceptance rules; Facilities owns leak and cleaning records; the service manager owns observations; the contractor owns load outcomes. Use current station mappings, incident tickets, service events and contractor codes. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Sensor or observation evidence applies only to the named cycle. A verified leak or contractor rejection outranks an indirect moisture indication. If environmental conditions or load linkage make the indication ambiguous, keep the event as possible wetting.

ANALYSIS REQUIRED

Operation: comparison and bounded diagnosis. Decision interval: weekly and after each wet-load rejection. Minimum evidence: six traceable cycles or a current serious leak, valid environmental context and at least one direct external observation or contractor outcome. Method: compare wet-condition frequency with service timing, station features and comparable dry points; rank supported operational explanations without asserting source behaviour. Uncertainty: show ambient-condition sensitivity, load-linkage confidence, unobserved contents and missing cycles. Safe failure: treat the issue as unconfirmed, address any current external leak under existing procedures and collect targeted cycle evidence.

ANSWER BOUNDARY

Do not inspect bags or readable waste, identify who deposited liquids, classify an unknown substance or instruct staff to contact or handle wet material. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-028

STAKEHOLDER QUESTION

Which contamination alerts may be false because of stale mappings, sensor faults, duplicate events or weak load-to-station linkage?

WHY THIS IS USEFUL

Prevents wasted response effort and protects stations from being labelled as problematic when the evidence chain is broken.

SENSORS AND TELEMETRY

Modality and location: Sensor-health records, station and load identifiers, service-reset scans, event deduplication keys, gateway receipt times and contractor traceability fields; no new sensing of waste contents. Coverage, timing and aggregation: Cover every contamination alert before operational escalation. Evaluate the full event chain from station mapping through service and contractor outcome; aggregate alert-quality flags and a supported, suspect or unavailable evidence state. Measurement discipline: Check identifier uniqueness, clock order, mapping validity, duplicate hashes, health status and the permitted many-to-one relationship between stations and shared loads. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The Estates waste lead owns station mappings; the service manager owns event records; the sensor service owner owns health data; the contractor owns load identifiers and outcomes. Use current mapping versions, signed service events, integration logs and original contractor records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Quality assessment expires when any source record changes. Original signed or source-system events outrank transformed dashboard rows. Where two sources claim different load links or times, retain both, stop attribution and assign the conflict to their owners.

ANALYSIS REQUIRED

Operation: diagnosis of evidence quality. Decision interval: before every contamination escalation and in a daily exception sweep. Minimum evidence: original alert, current mappings, relevant health and timing data, unique event identifiers and traceable service or load linkage. Method: test a fixed set of failure modes, calculate evidence-chain completeness and classify the alert as supported, suspect, duplicate or unavailable. Uncertainty: show each failed link, clock offset, stale mapping and alternative valid interpretation. Safe failure: suppress station attribution, preserve the raw event and request source-owner correction or a current authorised observation.

ANSWER BOUNDARY

Do not delete inconvenient source evidence, infer that a person or team caused the alert, inspect contents to compensate for weak linkage or silently merge conflicting events. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

segregation, contamination and signage

QUESTIONS 09-10 OF 10

STAKEHOLDER QUESTION

WM-029

Can this contamination alert be escalated now, or must it remain an unverified observation pending better evidence?

WHY THIS IS USEFUL

Provides a clear actionability gate so the team responds consistently without overclaiming what a single alert proves.

SENSORS AND TELEMETRY

Modality and location: The existing external observation, contractor code, stream and station identifiers, service event and relevant sensor-health evidence; no additional content inspection is required by the service. Coverage, timing and aggregation: Cover the single alert and its direct evidence chain. Capture each source at its original event time and aggregate a checklist of identity, timing, stream rule, observation scope, traceability and health sufficiency. Measurement discipline: Use the current evidence checklist, verify source timestamps and signatures or device health and record explicit not-observed items rather than assuming them. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability owns the escalation criteria; the service manager owns operational events; the contractor owns load outcomes; the Estates waste lead owns mappings. Use the effective escalation rule, original observations, current mappings and source evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The decision applies only to the named alert and expires if source evidence changes. A current competent-person direction overrides the checklist. Missing or conflicting identity, time, stream or linkage information forces an unverified result.

ANALYSIS REQUIRED

Operation: lookup and deterministic evidence-gate calculation. Decision interval: at alert receipt and after any new supporting evidence arrives. Minimum evidence: current escalation criteria and the required minimum fields for identity, timing, stream, observation scope and traceability. Method: evaluate each criterion explicitly, return actionable, unverified or conflicting and list the unmet evidence items. Uncertainty: show that passing the gate confirms evidence sufficiency, not material composition, intent, responsibility or regulatory compliance. Safe failure: retain the alert as unverified, take only standing precautionary actions and seek an authorised observation or source clarification.

ANSWER BOUNDARY

Do not open bags or devices to obtain evidence, identify or blame a depositor, classify unknown material or present an unverified alert as a confirmed contamination incident. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

STAKEHOLDER QUESTION

WM-030

Which single signage, aperture, station-pairing or placement change should be tested next, and what evidence would show whether it helped?

WHY THIS IS USEFUL

Turns repeated segregation problems into a controlled, measurable local test with a clear stop rule and owner decision point.

SENSORS AND TELEMETRY

Modality and location: Station configuration records, coded ordinary-contamination events, service resets, stream weights and external sign or aperture checks before and after the authorised change; privacy-preserving zone totals may be context only. Coverage, timing and aggregation: Cover one defined intervention station and a matched comparison or stable pre-period for at least six cycles before and six after where practical. Aggregate contamination outcomes, service demand and unintended overflow or access effects by cycle. Measurement discipline: Lock the intervention date and configuration, verify mapping continuity, use the same outcome definition across periods and exclude unhealthy or untraceable cycles. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability owns the segregation objective; the Estates waste lead owns station configuration; Accessibility, Fire Safety and the local owner control placement constraints; the service manager owns operations. Use an authorised test plan, current configuration and traceable outcome records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The result applies only to the tested configuration and context. Current safety or accessibility direction can stop the test. If concurrent service or stream changes prevent attribution, report the outcome as inconclusive rather than assigning effect to the intervention.

ANALYSIS REQUIRED

Operation: recommendation followed by before-after comparison. Decision interval: at test selection, each agreed checkpoint and test close. Minimum evidence: a supported problem, one feasible authorised intervention, stable outcome definition, baseline evidence and a pre-stated comparison and stop rule. Method: rank feasible options by evidence and constraints, select one for owner decision, then compare outcome ranges and unintended effects against the pre-stated rule. Uncertainty: show sample size, concurrent changes, comparison quality and whether the result supports direction, magnitude or neither. Safe failure: do not claim success or failure; retain the prior authorised arrangement or seek a better-controlled test.

ANSWER BOUNDARY

Do not manipulate users, collect person-level behaviour, claim a causal effect without a suitable design or implement, continue or stop a physical change without the authorised owners. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

hazardous, confidential and specialist waste

QUESTIONS 01-02 OF 10

WM-031

STAKEHOLDER QUESTION

Which approved battery, lamp, toner, small-WEEE or other specialist containers are ready for an authorised collection under their current acceptance rules?

WHY THIS IS USEFUL

Allows the team to book the correct specialist service only when the container, stream and acceptance conditions are supported.

SENSORS AND TELEMETRY

Modality and location: External load cells or manufacturer fill indicators where approved, container door or seal state, asset-tag scans and a coded external condition check at the specialist container; no opening, imaging or item-level content recognition. Coverage, timing and aggregation: Cover only registered specialist containers and their authorised stores. Sample integral devices at the supported rate and perform a current external check before booking; aggregate fill or weight range, seal, condition, storage age and collection-readiness fields per container. Measurement discipline: Validate load or fill against a known empty state, function-test door and seal inputs and treat density-dependent fill as a range rather than an exact item count. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup, observation and readiness calculation. Decision interval: before every specialist collection booking and at each authorised store check. Minimum evidence: registered container and stream, current external state, effective acceptance criteria and an authorised collection route. Method: test each required condition separately and return ready, not ready or unknown with the failed or missing conditions. Uncertainty: show weight or fill tolerance, unobserved internal state, evidence age and any contractor-specific assumption. Safe failure: do not book from analytics; keep the container under current storage controls and request a competent external check.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The specialist-waste owner owns acceptance and booking criteria; the Estates waste lead owns container records; Health and Safety owns storage controls; the contractor owns service-specific acceptance. Use current procedures, container specifications, service contract and authorised booking records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Readiness expires after any deposit, seal event, condition change or acceptance-rule change. Competent-person restrictions and contractor acceptance rules both must be satisfied. If they conflict, return not ready and refer the discrepancy to the named owners.

ANSWER BOUNDARY

Do not open the container, count or identify discarded items, infer chemistry or battery type from weight, or authorise packing, handling, transport or carrier selection. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-032

STAKEHOLDER QUESTION

Which confidential-paper consoles are due for an authorised collection without inspecting or imaging their contents?

WHY THIS IS USEFUL

Supports timely secure collection while preserving confidentiality and limiting access to need-to-know operational status.

SENSORS AND TELEMETRY

Modality and location: External load cells or approved non-imaging fill sensors fitted to the console, door or lock state and authorised service scans at the console; no camera, optical character recognition or readable-content capture. Coverage, timing and aggregation: Cover registered confidential consoles only. Sample fill at 15-minute or slower intervals appropriate to the device and capture service events at access and collection; aggregate threshold band, lock state, last collection and evidence-health state per console. Measurement discipline: Calibrate against known empty and reference loads, detect door-open periods that invalidate fill interpretation and keep console location and status access-controlled. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: observation, threshold calculation and recommendation. Decision interval: daily for authorised service planning and before each secure collection. Minimum evidence: console identity, current authorised threshold, current lock or door state, last secure service and a valid fill measure or authorised external observation. Method: calculate threshold band and schedule margin, then list consoles due, approaching due or unavailable without revealing detailed location to unauthorised roles. Uncertainty: show measurement error, door-open invalidation, evidence age and whether location detail has been withheld. Safe failure: retain the approved fixed schedule or request an authorised secure-service check; do not infer contents.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Records Management or Information Governance owns confidential-waste procedure; the Estates waste lead owns console records; Security or Access Control owns authorised access state; the secure-destruction contractor owns collection and destruction events. Use current service schedules, console register and signed secure-service records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A due state expires after an authorised service or threshold-changing deposit. Current information-governance direction outranks fill optimisation. If sensor state, lock state and service record conflict, expose a restricted exception and require owner confirmation.

ANSWER BOUNDARY

Do not open, image, read, sample or describe documents, disclose restricted console locations or access histories, or claim that destruction has occurred before the authorised certificate is received. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

hazardous, confidential and specialist waste

QUESTIONS 03-04 OF 10

WM-033

STAKEHOLDER QUESTION

Which authorised sharps or clinical-waste points require competent attention based only on their external status?

WHY THIS IS USEFUL

Ensures a full, damaged or incorrectly secured point reaches the competent owner without inviting untrained inspection or handling.

SENSORS AND TELEMETRY

Modality and location: Manufacturer-integrated fill indicator or external load cell where approved, lid or lock state, tamper indicator and a coded external condition report at the registered point; no opening or visual examination of contents. Coverage, timing and aggregation: Cover only points that the competent clinical or safety owner has registered. Sample integral status at the supported interval and capture a current external check on the prescribed schedule or after an alert; aggregate full-band, closure, damage and evidence-health states. Measurement discipline: Use only manufacturer-approved indicators, function-test closure inputs without opening the container and treat weight as a threshold range, not an item count or material classification. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: observation and escalation recommendation. Decision interval: at prescribed checks, on any alert and before authorised collection. Minimum evidence: registered point, current external status, effective competent procedure and current response contact route. Method: apply only the approved external-state thresholds and rank the required contact urgency without diagnosing contents or exposure. Uncertainty: show the observable limit, measurement tolerance, last valid status and any unavailable device or access evidence. Safe failure: return competent attention required or status unknown, keep existing controls in place and use the approved escalation route.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The competent clinical or Health and Safety owner controls the stream and response; the Estates waste lead owns the location register; the approved specialist contractor owns collection acceptance. Use current clinical-waste procedures, container instructions and authorised service records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: External-status evidence expires after any deposit, closure event, movement or reported damage. Competent-person direction outranks all telemetry. A suspected device fault or conflicting state requires a restricted exception and no normal-status answer.

ANSWER BOUNDARY

Do not touch, open, re-close, move or inspect the container, estimate exposure, identify discarded items or instruct anyone without the named competence and authority. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-034

STAKEHOLDER QUESTION

Which WEEE or data-bearing items have an authorised release and complete chain-of-custody evidence ready for secure collection?

WHY THIS IS USEFUL

Prevents an unreleased or weakly traceable asset from leaving the building and supports a clean secure handover.

SENSORS AND TELEMETRY

Modality and location: Asset or cage tag scans, seal and door state, authorised release status from the asset system and handover scans at the secure store and carrier interface; no access to device data or files. Coverage, timing and aggregation: Cover each item or sealed batch named for collection from authorised release through carrier handover. Capture every custody event and aggregate release, location, seal, custodian-role and next authorised handover state per item or batch. Measurement discipline: Verify tag uniqueness, enforce ordered event times, detect seal changes and use scanner location only as a controlled handover point, not person tracking. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and chain-of-custody reconciliation. Decision interval: before movement to secure storage, before carrier handover and after receipt. Minimum evidence: unique item or batch ID, current authorised release, required security status, ordered custody events and approved carrier booking. Method: validate release, route and event sequence, detect gaps or duplicate IDs and return ready, held or unresolved for each item or batch. Uncertainty: show unobserved intervals, seal-state uncertainty, role-restricted fields and any many-item batch assumption. Safe failure: hold the item under current secure controls and request the relevant asset or information-security owner to resolve the gap.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The asset owner and Finance or Procurement own release; Information Security owns data-bearing controls; the Estates waste lead owns waste routing; Security owns store access; the approved contractor owns carrier handover. Use current asset-disposal authorisation, secure-store records, custody events and carrier documentation. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Release expires if the asset status, seal or disposal route changes. Information Security and asset-owner restrictions outrank collection convenience. Missing, out-of-order or conflicting custody events block ready status and remain visible.

ANSWER BOUNDARY

Do not access, test or erase device data, reveal sensitive asset details broadly, infer data sanitisation from collection status or authorise release, transport or destruction. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

hazardous, confidential and specialist waste

QUESTIONS 05-06 OF 10

WM-035

STAKEHOLDER QUESTION

Is any approved battery-waste store showing abnormal heat, damage, smoke or alarm conditions that require emergency escalation?

WHY THIS IS USEFUL

Gives the team a rapid route to the current emergency response without asking them to classify batteries or manage the incident themselves.

SENSORS AND TELEMETRY

Modality and location: Building fire and smoke alarms as the primary alarm source, approved fixed temperature sensors or thermal probes in the store, external door status and authorised external condition reports; no opening or thermal imaging of individual discarded items by the service. Coverage, timing and aggregation: Cover only approved battery stores with verified alarm mapping. Sample temperature at the commissioned rate, preserve raw alarm transitions and aggregate current alarm, temperature range, rate of rise, door state and evidence health for the store. Measurement discipline: Use fire-system tests and calibrated temperature references, verify alarm-zone mapping and treat a single unverified sensor anomaly as uncertain while preserving any credible external report. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: observation and emergency escalation routing. Decision interval: continuous for commissioned alarms and immediately on any credible report. Minimum evidence: verified store mapping and either an authoritative alarm, a valid abnormal temperature state or a credible current external report. Method: apply the current emergency trigger rules, preserve source times and route the alert to the authorised emergency channel without diagnosing the material. Uncertainty: show sensor or mapping health, alarm source and any uncertainty only after escalation has been initiated by the proper system or person. Safe failure: treat the situation as potentially serious, follow the current emergency instruction and provide no handling advice.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Fire Safety and the emergency controller own alarm interpretation and response; Health and Safety owns battery-storage controls; Facilities owns building systems; the Estates waste lead owns waste-store records. Use current alarm events, emergency instructions and competent-person directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Any active alarm or credible abnormal external report is current until the emergency owner clears it. Emergency direction outranks all other records. Conflicting normal telemetry must not suppress an alarm or report.

ANSWER BOUNDARY

Do not enter, open, move, cool, isolate or classify batteries, estimate fire risk, override alarms or substitute analytics for Fire Safety, emergency services or other competent direction. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-036

STAKEHOLDER QUESTION

Are spent lamps and other fragile specialist items externally reported as correctly packaged, contained and undamaged for the next authorised handover?

WHY THIS IS USEFUL

Helps the team identify a packaging or damage exception before a specialist carrier arrives, without disturbing fragile contents.

SENSORS AND TELEMETRY

Modality and location: Container seal state, tilt or shock indicator where approved and a coded external package check at the authorised store; no opening, shaking or internal image capture. Coverage, timing and aggregation: Cover each registered fragile-waste package or approved batch prepared for handover. Capture at packing completion, any movement, pre-collection check and carrier handover; aggregate seal, orientation, visible package condition, shock event and evidence age. Measurement discipline: Function-test discrete indicators, validate shock thresholds against the approved package specification and treat a missing indicator as unknown, not undamaged. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: observation and readiness recommendation. Decision interval: after packing, after movement and immediately before handover. Minimum evidence: registered package or batch, current packaging specification, external condition, seal and any required indicator state. Method: check every external requirement and return ready, not ready or unknown with the exact unmet condition. Uncertainty: show unobserved internal condition, device tolerance, movement gaps and any batch-level assumption. Safe failure: keep the package in its current controlled location and refer it to the competent owner; do not attempt correction.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The specialist-waste owner owns packaging instructions; Health and Safety owns handling controls; the Estates waste lead owns storage records; the contractor owns acceptance criteria. Use current package specifications, movement records and authorised external condition checks. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Condition evidence expires after movement, impact or seal change. Competent-person direction and current carrier acceptance rules both apply. If an indicator and external observation conflict, use the more conservative state and request competent confirmation.

ANSWER BOUNDARY

Do not open, re-pack, straighten, shake or handle the package, infer breakage or material release from weight alone, or authorise acceptance or transport. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

hazardous, confidential and specialist waste

QUESTIONS 07-08 OF 10

WM-037

STAKEHOLDER QUESTION

Which chemical, laboratory or other specialist-waste requests are waiting for classification or acceptance by the competent owner?

WHY THIS IS USEFUL

Gives the team a safe workflow list while making clear that sensors cannot determine composition or authorise a route.

SENSORS AND TELEMETRY

Modality and location: Workflow status, request identifiers, authorised container or area alarms and current storage-environment telemetry only where specified by the competent owner; no composition sensing, bag inspection or automated material recognition. Coverage, timing and aggregation: Cover each submitted specialist request from initial record to competent classification, acceptance or closure. Capture workflow events as they occur and environmental alarms at their supported rate; aggregate request age, missing fields, competent-owner queue and current precautionary status. Measurement discipline: Validate request identifiers and event order, keep environmental telemetry separate from material identity and treat absent composition data as intentionally unknown. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The originating laboratory or technical owner supplies the request; the competent hazardous-waste or Health and Safety owner controls classification and acceptance; the Estates waste lead manages approved logistics. Use current request forms, safety information supplied by authorised owners and competent decisions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Workflow status is current until a new signed event; safety information follows its effective version. Competent decisions outrank request descriptions and telemetry. Conflicting or incomplete material information keeps the request pending and prevents route assignment.

ANALYSIS REQUIRED

Operation: lookup and workflow-state calculation. Decision interval: daily and whenever a new specialist request or competent decision arrives. Minimum evidence: unique request, accountable originator, current competent owner, required information checklist and latest signed workflow event. Method: calculate pending stage, age and missing authorised information, then group requests by competent owner and urgency rule without classifying material. Uncertainty: show missing fields, stale safety information, disputed ownership and any environmental alarm that requires separate action. Safe failure: return classification or acceptance pending and use the current precautionary procedure; assign no waste code or route.

ANSWER BOUNDARY

Do not infer composition from location, container, sensor or request wording, invent a waste code, advise handling or transport, or disclose sensitive research or chemical details beyond authorised roles. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-038

STAKEHOLDER QUESTION

Are specialist-waste containers within their breached storage-time, fill, seal, segregation and environmental limits?

WHY THIS IS USEFUL

Supports a clear exception list for the competent owner before a limit is exceeded or a collection is booked incorrectly.

SENSORS AND TELEMETRY

Modality and location: Container or store tag scans, approved fill or weight indicator, seal and door state, temperature or humidity sensing only where the written procedure specifies it, and coded external segregation checks. Coverage, timing and aggregation: Cover every registered specialist container and required store zone. Sample environmental quantities at the commissioned rate, capture deposits or seal events where authorised and perform checks on the prescribed schedule; aggregate each limit and evidence-health state separately. Measurement discipline: Use procedure-specific calibration ranges, verify sensor placement represents the controlled zone and never infer compatibility or material stability from temperature or humidity alone. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The competent specialist-waste owner owns limits and segregation rules; Health and Safety owns storage controls; Facilities owns environmental systems; the Estates waste lead owns container records; the contractor owns acceptance. Use current procedures, container records and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Each state expires at its next deposit, seal, movement or environmental sample, and storage age uses the authoritative start event. Competent-person and emergency direction outrank calculations. If the start time, stream or limit is disputed, return unknown and escalate.

ANALYSIS REQUIRED

Operation: lookup, observation and multi-limit calculation. Decision interval: at the prescribed check interval and before collection or movement. Minimum evidence: registered container, authoritative storage start, current effective limits and valid current state for every applicable dimension. Method: calculate margin to each limit independently, identify breached, approaching, within or unknown and avoid combining them into a misleading single score. Uncertainty: show sensor tolerance, coverage location, unobserved internal state and missing deposit or seal events. Safe failure: apply the current precautionary status and refer the container to the competent owner; do not infer that limits are satisfied.

ANSWER BOUNDARY

Do not classify contents, open or sample containers, claim safe compatibility, extend storage time or authorise segregation, movement, collection or transport. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

hazardous, confidential and specialist waste

QUESTIONS 09-10 OF 10

WM-039

STAKEHOLDER QUESTION

Are specialist-waste movements traceable from authorised release through storage and carrier handover to the recorded receiving destination?

WHY THIS IS USEFUL

Allows the team to locate a documentary custody gap before reporting or payment and to preserve a defensible movement history.

SENSORS AND TELEMETRY

Modality and location: Barcode or RFID scans at authorised handover points, seal state and signed digital custody events linked to the container, consignment or batch; vehicle data are limited to the booked movement event, not continuous tracking. Coverage, timing and aggregation: Cover the full movement chain for each specialist consignment. Capture release, store entry, store exit, carrier acceptance and receiving evidence; aggregate ordered events, custody role, seal state, timestamps and unresolved gaps per consignment. Measurement discipline: Verify identifier uniqueness, clock sequence and expected handover points, detect duplicate or reversed events and preserve original signed records. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and chain-of-custody reconciliation. Decision interval: at each handover and before any movement is reported as complete. Minimum evidence: unique consignment or batch, authorised release, required custody stages, original signed events and receiving destination record. Method: order events, test required transitions, identify missing, duplicate, out-of-order or conflicting custody states and return complete, incomplete or unresolved. Uncertainty: show unobserved transit, clock differences, batch aggregation and any role-restricted fields. Safe failure: leave the movement incomplete, preserve all source records and refer the exact gap to the responsible custody owner.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The competent waste owner controls release; Security or store owners control custody points; the carrier owns acceptance; the receiving facility owns receipt; the Estates waste lead owns the movement register. Use authorised consignment records, signed handovers and receiving evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Traceability remains open until the required receiving evidence is accepted by the owner. Original signed events outrank copied dashboard entries. A missing or conflicting handover prevents a complete-chain answer and must not be filled by assumed transit.

ANSWER BOUNDARY

Do not disclose restricted store or route details broadly, identify individual workers where role-level custody is sufficient, infer receipt from departure or authorise movement or transport. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-040

STAKEHOLDER QUESTION

What is the safest permitted next step when an unknown, unlabelled, damaged or leaking specialist item is reported?

WHY THIS IS USEFUL

Directs the team to the current precautionary and competent response without attempting identification or creating unsafe handling instructions.

SENSORS AND TELEMETRY

Modality and location: The original authorised external report, area alarm or leak detector, current location or zone record and contact-status telemetry for the responsible emergency or specialist owner; sensing is not used to identify the material. Coverage, timing and aggregation: Cover only the reported area and current alert. Preserve the first report, subsequent authorised observations and alarm times; aggregate location confidence, external condition, active alarms, access restriction and responsible contact route. Measurement discipline: Validate location and alarm mapping, retain exact source wording, distinguish observed leakage from device alarm and do not convert colour, smell, container shape or sensor readings into composition. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and precautionary escalation routing. Decision interval: immediately on report and at each authorised status update. Minimum evidence: a credible report or alarm, current area identity, effective unknown-material procedure and an authorised emergency or competent contact route. Method: return only the approved first-response instruction for the role, preserve uncertainty and route the case to the competent owner. Uncertainty: show what was reported versus sensed, location confidence and any unavailable contact or access information. Safe failure: treat the material as unknown, follow current precautionary instructions and avoid all handling advice beyond the authorised procedure.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The emergency controller, Health and Safety and the competent specialist owner control the response; Security and the building manager control access restrictions. Use current emergency and unknown-material procedures, alarms and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The latest competent or emergency direction is immediately authoritative. If location, condition or owner is uncertain, use the most precautionary current procedure. Conflicting normal telemetry never clears a credible report.

ANSWER BOUNDARY

Do not approach, touch, open, move, smell, sample, photograph for classification or identify the item, estimate exposure, or authorise isolation, handling, transport or disposal. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

routes, lifts, loading areas and access

WM-041

STAKEHOLDER QUESTION

Which approved internal route is authorised and usable now for the planned bin, cage or trolley movement?

WHY THIS IS USEFUL

Lets the team select a currently valid route and avoid arriving at a closed door, blocked corridor or unauthorised zone.

SENSORS AND TELEMETRY

Modality and location: Current door and gate status, lift service state, coded obstruction reports and route-point checks at authorised waypoints; no person tracking and no reliance on the May 2021 plan for current use. Coverage, timing and aggregation: Cover only routes in the current approved route register. Refresh integrated status at one to five minute intervals and perform a route check immediately before movement; aggregate each segment as open, restricted, closed, obstructed or unknown. Measurement discipline: Function-test discrete states, verify waypoint and door mapping, time-align route events and treat missing access or obstruction evidence as unknown rather than open. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup, observation and path-feasibility calculation. Decision interval: before every planned movement and at any route-status change. Minimum evidence: current authorised route graph, movement origin and destination, required access state and current status for every segment. Method: exclude closed, restricted or unknown segments, test continuity and return the shortest authorised feasible route or no confirmed route. Uncertainty: show unverified segments, status age, mapping confidence and any access detail withheld by role. Safe failure: do not route the movement; use the standing contingency and ask the current route owner for confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The building manager and Facilities own current route and closure records; Security or Access Control owns access state; Fire Safety owns emergency restrictions; the waste service manager owns movement plans. Use current route authorisations, closures, work notices and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Route state expires at the movement start or any status change. Emergency, fire and security direction outrank bookings and telemetry. The May 2021 floor plan is historical only. If segment records conflict, the route is not confirmed.

ANSWER BOUNDARY

Do not infer access from the historical plan, disclose sensitive access rules, direct entry through a restricted area or unlock, hold or control any door, gate or lift. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-042

STAKEHOLDER QUESTION

Are route widths, turns, thresholds, doors, slopes and temporary obstructions compatible with the specified trolley and load?

WHY THIS IS USEFUL

Prevents a crew from committing to a route that cannot physically accommodate the planned equipment and load envelope.

SENSORS AND TELEMETRY

Modality and location: Current surveyed dimensions, QR-linked route-point measurements, door-open clearance, slope and threshold measurements, trolley dimensions and coded temporary-obstruction checks; no inference from historic drawings alone. Coverage, timing and aggregation: Cover every critical segment on the proposed route and the exact trolley or cage configuration. Reconfirm fixed dimensions after works and temporary clearances immediately before movement; aggregate minimum clearance and failed constraint per segment. Measurement discipline: Use calibrated distance and slope tools, record measurement tolerance and door state, and verify that the measured path represents the usable envelope rather than architectural nominal width. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: calculation and feasibility comparison. Decision interval: during movement planning and immediately before execution. Minimum evidence: current usable route dimensions, exact trolley and load envelope, required safety clearance and a current obstruction state. Method: calculate clearance margins for width, height, turning, threshold and slope at every segment and identify the limiting point. Uncertainty: show measurement tolerance, load-shape uncertainty, door-state dependency and unmeasured segments. Safe failure: reject the route as unconfirmed and request a current measured check or an authorised alternative.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Facilities or Space Management owns current surveyed geometry; the building manager owns temporary obstructions; the equipment owner owns trolley dimensions; Health and Safety owns movement constraints. Use current measured route records, equipment specifications and local restrictions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Temporary-clearance evidence expires at movement start; fixed geometry becomes stale after works or layout changes. Competent manual-handling or safety restrictions outrank dimensional fit. If measurement tolerance overlaps a required clearance, return not confirmed.

ANSWER BOUNDARY

Do not declare manual handling safe, instruct staff to force equipment through a marginal gap, infer geometry from the May 2021 plan or authorise route changes. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

routes, lifts, loading areas and access

QUESTIONS 03-04 OF 10

STAKEHOLDER QUESTION

WM-043

Which lift, if any, is authorised, available, large enough and prepared for the planned waste movement?

WHY THIS IS USEFUL

Helps the team book the correct lift and avoid unsafe loading, service disruption or an unusable car.

SENSORS AND TELEMETRY

Modality and location: Lift service and fault status, booking state, car and door dimensions, rated load, protective-cover readiness and authorised route checks at the lift lobbies; no control of lift operation. Coverage, timing and aggregation: Cover lifts listed in the current waste-movement procedure and the route to both lobbies. Refresh service state at one-minute intervals where integrated and confirm booking, protection and access before movement; aggregate suitability by lift. Measurement discipline: Use certified current dimensions and rated load records, function-test status interfaces and do not infer usable payload from rated load without accounting for trolley and people as required by procedure. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and multi-constraint feasibility calculation. Decision interval: during booking, one hour before movement and immediately before entry. Minimum evidence: current authorised lift list, valid service state, dimensions, rated capacity, load envelope, booking and required protection. Method: test access, dimensions, payload, booking and preparation separately and return suitable, unsuitable or unknown for each lift. Uncertainty: show capacity margin, dimension tolerance, service-data age and any dependency on escorts or protection. Safe failure: use no lift recommendation and refer to the lift and waste service owners for a current plan.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The lift service owner owns availability and impairment; Facilities owns dimensions and protection arrangements; Security owns access; Health and Safety owns movement limits; the waste service manager owns booking. Use current lift records, bookings, procedures and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Availability expires at movement start or on any lift fault. Emergency recall, maintenance isolation and competent restrictions outrank bookings. Conflicting capacity or service states result in no confirmed lift.

ANSWER BOUNDARY

Do not call, reserve, isolate, hold or control a lift, override a fault or emergency state, or declare loading or manual handling safe. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

STAKEHOLDER QUESTION

WM-044

Which authorised movement window is likely to minimise conflict with teaching, public circulation, deliveries and cleaning activity?

WHY THIS IS USEFUL

Supports a practical time-window proposal that reduces congestion without tracking individuals or promising an interruption-free route.

SENSORS AND TELEMETRY

Modality and location: Approved timetables and bookings, privacy-preserving aggregate people counts at route zones where authorised, loading-bay and delivery bookings, lift bookings and cleaning schedules. Coverage, timing and aggregation: Cover the approved route and candidate windows for the next operational day or event. Aggregate zone counts to at least 15-minute intervals and combine only with current bookings and closures; no device or individual trajectory data. Measurement discipline: Validate counter coverage and direction accuracy, mark sensor outages, align all schedules to the same local time and distinguish planned activity from observed aggregate presence. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: forecast, comparison and recommendation. Decision interval: for the next day, confirmed event or booked movement. Minimum evidence: at least two feasible authorised windows, current schedules and restrictions, and valid aggregate evidence for any congestion claim. Method: score each window against route closures, bookings and observed aggregate load, present the best-supported options and list conflicts that remain. Uncertainty: show counter coverage, schedule uncertainty, unobserved zones and that low counts do not guarantee a clear route. Safe failure: offer no preferred window and use the current authorised default or seek building-manager confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Timetable and event owners supply planned use; the building manager owns local restrictions; Security owns access windows; Facilities owns delivery and lift bookings; the service manager owns movement constraints. Use current effective schedules and verified aggregate observations. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A proposed window expires when a booking, closure or access condition changes. Emergency and current operational directions outrank low aggregate counts. If count coverage is incomplete, present only schedule-based options with reduced confidence.

ANSWER BOUNDARY

Do not identify or follow people, infer class attendance, expose sensitive schedules broadly, guarantee an empty route or book staff, lifts or loading areas. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

routes, lifts, loading areas and access

QUESTIONS 05-06 OF 10

WM-045

STAKEHOLDER QUESTION

Are all required access permissions, keys, escorts, shutters and controlled-door arrangements ready for the planned movement?

WHY THIS IS USEFUL

Allows the team to resolve a missing access dependency before the crew and load reach a controlled boundary.

SENSORS AND TELEMETRY

Modality and location: Role-level access readiness from the authorised access system, key or escort booking status, shutter or door service state and coded pre-movement checks; no disclosure of individual access histories. Coverage, timing and aggregation: Cover each controlled boundary on the approved route and each required role or resource. Refresh system state immediately before movement and aggregate ready, missing, expired, faulted or unknown dependencies. Measurement discipline: Verify route-to-door mapping, role validity period and booking identity, function-test door-state interfaces and separate permission from physical door availability. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and dependency reconciliation. Decision interval: at planning, one hour before and immediately before movement. Minimum evidence: approved route, required role-level permissions, key or escort booking and current physical state for every controlled boundary. Method: evaluate each dependency, preserve separation between authorisation and equipment state and provide an exception list to authorised coordinators. Uncertainty: show expiry times, role-based redactions, device faults and any boundary that has not been tested. Safe failure: do not start the movement; refer missing dependencies to Security, Facilities or the building manager.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Security or Access Control owns permissions, keys and controlled-door state; the building manager owns escorts and local arrangements; Facilities owns shutters; the waste service manager owns movement requirements. Use current access approvals, bookings, fault records and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Readiness expires at the movement start or any access change. Security and emergency directions outrank bookings. If permission status cannot be exposed to the requesting role, return a restricted ready or not-ready result without sensitive details.

ANSWER BOUNDARY

Do not expose named-person access logs, create permissions, issue keys, nominate an escort, bypass controls or unlock, hold or control a door or shutter. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-046

STAKEHOLDER QUESTION

Can the booked collection vehicle approach, manoeuvre, exchange the container and depart under current site controls?

WHY THIS IS USEFUL

Reduces failed arrivals and unsafe improvisation by checking the current vehicle envelope, access and exchange conditions before the slot.

SENSORS AND TELEMETRY

Modality and location: Vehicle booking dimensions, current gate and yard status, bay occupancy sensing, height or width clearance records, external container position and a coded manoeuvre-area check; no continuous vehicle or driver tracking. Coverage, timing and aggregation: Cover the authorised approach, gate, manoeuvre area and exchange point for the booked slot. Refresh occupancy and gate state every one to five minutes and confirm clearances before arrival; aggregate each movement phase as feasible, blocked or unknown. Measurement discipline: Use current measured clearances, validate occupancy sensors against a physical reference, record weather or works only when authoritative and do not infer safe manoeuvring from geometric fit alone. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and staged feasibility calculation. Decision interval: at booking confirmation, two hours before and immediately before arrival. Minimum evidence: confirmed vehicle and exchange method, current authorised route, measured clearances, current bay or yard state and active site controls. Method: test approach, gate, manoeuvre, exchange and exit constraints separately and list the first blocking dependency. Uncertainty: show measurement margins, occupancy age, unobserved hazards and any reliance on contractor-supplied dimensions. Safe failure: hold the ready status and ask the site logistics owner and contractor to confirm a compliant plan.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The site logistics or building manager owns vehicle access; Security owns gates; Facilities owns yard condition and clearances; the contractor owns vehicle specifications; Fire Safety owns emergency restrictions. Use current booking, vehicle data, site controls and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Feasibility expires at the booked slot or any occupancy, closure or vehicle change. Current site and emergency controls outrank the booking. If vehicle dimensions or exchange method differ from the booking, return not confirmed.

ANSWER BOUNDARY

Do not direct the driver, certify a manoeuvre as safe, open gates, move parked vehicles or containers, or override traffic, fire or security controls. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

routes, lifts, loading areas and access

QUESTIONS 07-08 OF 10

WM-047

STAKEHOLDER QUESTION

Is the loading or exchange area clear, booked and operationally ready for the next confirmed collection?

WHY THIS IS USEFUL

Provides a simple go or not-ready status so the team can clear dependencies before the contractor reaches the area.

SENSORS AND TELEMETRY

Modality and location: Bay-booking status, ground-loop or overhead occupancy sensing where commissioned, gate and lighting state, coded obstruction and surface-condition check, and container readiness at the exchange point. Coverage, timing and aggregation: Cover the named bay or exchange area and its immediate safe operating envelope. Refresh occupancy and access at one to five minute intervals and complete a current external readiness check before the slot; aggregate readiness by required condition. Measurement discipline: Validate occupancy detection for the bay geometry, function-test gate and lighting states and treat blind zones or weather-obscured sensors as unknown. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The site logistics or building manager owns the bay and booking; Facilities owns surface, lighting and defects; Security owns access; the waste service manager owns container readiness; the contractor owns exchange requirements. Use current bookings, defect records and site instructions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Readiness expires at slot start or any new occupancy, obstruction, weather or defect event. Emergency and site controls outrank bookings. If the area is reported clear but a valid occupancy sensor shows occupied, use not ready until checked.

ANALYSIS REQUIRED

Operation: observation and readiness calculation. Decision interval: one hour before and immediately before the confirmed slot. Minimum evidence: current booking, bay identity, access state, occupancy or authorised observation, surface and lighting status and prepared container. Method: evaluate each required condition independently and return ready, not ready or unknown with the blocking item. Uncertainty: show blind zones, status age, weather effects and any condition that depends on a manual check. Safe failure: return not ready and notify the authorised site coordinator; do not call the area clear.

ANSWER BOUNDARY

Do not move obstructions or containers, direct traffic, open a gate, guarantee the area will remain clear or authorise collection outside the booked controls. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-048

STAKEHOLDER QUESTION

Which authorised alternative route is currently available when the normal route, lift or loading area is closed?

WHY THIS IS USEFUL

Lets the team activate a documented contingency option rather than improvising through an unverified space.

SENSORS AND TELEMETRY

Modality and location: Current closure and access records, route-segment status, lift availability, measured geometry and coded obstruction checks for pre-approved alternatives; no historic-plan inference or person tracking. Coverage, timing and aggregation: Cover only alternative routes authorised in the current contingency record. Refresh integrated states at one to five minute intervals and perform a current critical-point check before use; aggregate feasibility and unresolved dependencies by alternative. Measurement discipline: Verify mapping and dimensions for each alternative, time-align status events and treat any unmeasured or access-restricted segment as unknown. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The building manager owns contingency routes; Facilities owns route and lift state; Security owns access; Fire Safety owns emergency restrictions; Health and Safety owns handling constraints. Use current contingency records, closures and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: An alternative is current only for the stated closure and movement. Emergency and security controls outrank contingency preferences. The May 2021 floor plan cannot establish an alternative route. Conflicting segment states block selection.

ANALYSIS REQUIRED

Operation: lookup, comparison and path-feasibility calculation. Decision interval: when the normal route becomes unavailable and immediately before contingency use. Minimum evidence: current closure, at least one authorised alternative, complete current segment status, dimensions and required permissions. Method: exclude unavailable segments, test each alternative against movement and access constraints and present supported options in owner-defined priority order. Uncertainty: show status age, unverified segments, capacity margins and any dependency on temporary permissions or equipment. Safe failure: state that no confirmed alternative exists and refer the movement to the current contingency owner.

ANSWER BOUNDARY

Do not create an unofficial route, direct passage through a restricted or fire-protected area, infer current use from the historical plan or control access equipment. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

routes, lifts, loading areas and access

QUESTIONS 09-10 OF 10

WM-049

STAKEHOLDER QUESTION

What load, trolley and manual-handling constraints need competent confirmation before the planned waste movement?

WHY THIS IS USEFUL

Ensures the movement request reaches a competent person with a complete evidence pack instead of producing unsafe automated advice.

SENSORS AND TELEMETRY

Modality and location: Verified load weight or bounded estimate, container dimensions, trolley specification, route geometry, lift capacity and coded external condition; no biomechanical sensing of staff and no automated handling instruction. Coverage, timing and aggregation: Cover the exact load, equipment and route proposed for the movement. Capture weight and dimensions before movement and update after any load or equipment change; aggregate the evidence fields and unresolved competent-decision points. Measurement discipline: Use calibrated scales or documented bounded estimation, verify trolley and container specifications and state when load distribution or centre of gravity is not observed. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Health and Safety or the competent manual-handling assessor owns handling decisions; the equipment owner owns trolley limits; Facilities owns route and lift data; the waste service manager owns the movement request. Use current risk controls, equipment data and measured route evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Evidence expires after a load, route, trolley or staffing change. Competent-person restrictions outrank calculated margins. If weight, dimensions or route are uncertain, no handling recommendation is produced.

ANALYSIS REQUIRED

Operation: calculation and evidence-pack preparation. Decision interval: during planning and immediately after any material change. Minimum evidence: current load range, container and trolley specification, route constraints and named competent decision owner. Method: calculate factual margins only, list unknowns and assemble the evidence required for the competent person to decide. Uncertainty: show weight tolerance, unknown load distribution, route measurement limits and any excluded handling factors. Safe failure: return competent confirmation required and keep the movement on hold under current procedure.

ANSWER BOUNDARY

Do not prescribe team size, posture, lifting technique or equipment use, declare manual handling safe or authorise the movement. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-050

STAKEHOLDER QUESTION

Where are route delays, container damage or near-miss reports recurring enough to require a route or movement-method assessment?

WHY THIS IS USEFUL

Focuses management attention on repeated route problems while avoiding blame and unsupported claims about why they occurred.

SENSORS AND TELEMETRY

Modality and location: Role-neutral route start and end events, segment-level delay codes, asset damage records, lift or door fault events and authorised near-miss reports; no person identity, wearable tracking or audio capture. Coverage, timing and aggregation: Cover at least eight comparable movements on each candidate route, with event data aggregated by route segment, equipment type and time window. Preserve serious single events separately from recurrence analysis. Measurement discipline: Validate event definitions, remove duplicate reports, align route and fault clocks and separate missing timing events from zero delay. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The waste service manager owns movement records; Facilities owns door, lift and route faults; Health and Safety owns near-miss records; the asset owner owns damage records. Use current route versions, incident records and equipment data. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Comparisons split after a route, equipment or procedure change. Serious competent-person direction applies immediately regardless of recurrence count. Where incident and timing evidence conflict, show both and do not assign cause.

ANALYSIS REQUIRED

Operation: comparison and bounded diagnosis. Decision interval: monthly and immediately after any serious report. Minimum evidence: eight comparable movements or a policy-defined serious event, current route version and traceable delay, damage or near-miss evidence. Method: calculate event rates and delay distributions by segment, compare with matched routes and rank supported route or equipment factors without attributing personal fault. Uncertainty: show under-reporting risk, missing events, changing conditions and that recurrence does not establish a single cause. Safe failure: state that recurrence is not established, preserve the event-level response and collect targeted route evidence.

ANSWER BOUNDARY

Do not identify or rank staff performance, infer unsafe behaviour, disclose sensitive incident details broadly or change routes, equipment or procedures without authorised owners. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

contractor coordination, storage and collections

QUESTIONS 01-02 OF 10

WM-051

STAKEHOLDER QUESTION

Which contractor collections are genuinely confirmed today, including stream, container, arrival window and operational contact route?

WHY THIS IS USEFUL

Creates a single reliable collection schedule and prevents crews preparing for an unconfirmed or incorrectly scoped visit.

SENSORS AND TELEMETRY

Modality and location: Authorised booking and contractor-confirmation events, container readiness status and site access or bay booking state; no driver-location tracking and no assumption from an automated calendar entry alone. Coverage, timing and aggregation: Cover every collection expected for the current day. Refresh booking and confirmation status whenever either party changes it and aggregate one state per collection: confirmed, provisional, changed, cancelled, conflicting or unknown. Measurement discipline: Validate unique booking IDs, source signatures, time zones and container mapping, and distinguish system delivery receipt from contractor acceptance. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and booking reconciliation. Decision interval: at start of day, two hours before each window and on every change event. Minimum evidence: unique booking, current contractor acceptance, current site acceptance, stream, container, arrival window and authorised contact route. Method: join site and contractor records, compare key fields and return confirmed, provisional, changed, cancelled, conflicting or unknown. Uncertainty: show source times, confirmation age, unresolved field differences and any contact detail hidden by permission. Safe failure: use the standing schedule as provisional only and contact the authorised coordinator; do not prepare a high-risk handover on an unknown booking.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The waste service manager owns the site schedule; Procurement or Contract Management owns the contract scope; the contractor owns its confirmation; the building or logistics manager owns site access and bay bookings. Use current signed bookings, change events and authorised contact records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Confirmation expires after any material change and at the end of the arrival window. A direct current contractor confirmation and site acceptance must both exist. If systems disagree on time, stream or container, expose the conflict and do not call the visit confirmed.

ANSWER BOUNDARY

Do not disclose personal phone numbers broadly, infer arrival from a calendar entry, contact a driver through unauthorised channels or book, cancel or alter a collection. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-052

STAKEHOLDER QUESTION

Has the contractor arrived, been delayed or missed the agreed window based on authorised site events?

WHY THIS IS USEFUL

Lets the team update storage and staffing plans using defensible arrival evidence without tracking a driver or vehicle continuously.

SENSORS AND TELEMETRY

Modality and location: Gate or reception arrival event, loading-bay occupancy, authorised contractor check-in and collection start or completion event linked to the booking; no continuous vehicle GPS or person tracking. Coverage, timing and aggregation: Cover the booked access and service points for the active window. Capture event times as they occur and aggregate not due, on site, in service, completed, delayed, missed or unknown for the booking. Measurement discipline: Verify event-to-booking linkage, clock order and duplicate check-ins, and treat bay occupancy without a matched booking as unidentified occupancy rather than contractor arrival. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: observation and time-window calculation. Decision interval: throughout the booked window and at its close. Minimum evidence: current booking window and at least one authoritative matched site or contractor event. Method: order matched events, compare them with the accepted window and return the current booking state with elapsed delay where supported. Uncertainty: show clock offset, unmatched occupancy, missing completion evidence and whether delay is observed or contractor-reported. Safe failure: return arrival unknown or window missed, preserve storage contingency and contact the authorised coordinator.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Security or reception owns authorised arrival events; the site logistics owner owns bay status; the contractor owns its delay notice; the waste service manager owns service start and completion records. Use original booking, authorised site events and contractor communications. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Status expires on the next booking or service event. The agreed window and current authorised variation define lateness. If site and contractor events conflict, show both and keep arrival state unresolved until a source owner confirms.

ANSWER BOUNDARY

Do not identify or track a driver, infer cause of delay, direct vehicle movement, change access controls or promise a new arrival time. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

contractor coordination, storage and collections

QUESTIONS 03-04 OF 10

WM-053

STAKEHOLDER QUESTION

Are the booking, site instructions, stream, container, vehicle and contact details consistent before the contractor arrives?

WHY THIS IS USEFUL

Catches scope and access mismatches early enough for the authorised coordinators to correct them before a failed collection.

SENSORS AND TELEMETRY

Modality and location: Structured booking records, current site-instruction version, container and stream identifiers, vehicle dimensions, access and bay requirements and authorised role-level contact routes; no waste-content sensing is needed. Coverage, timing and aggregation: Cover every confirmed or provisional booking within the next two working days. Recalculate after any source change and aggregate matched, mismatched, missing and restricted fields per booking. Measurement discipline: Validate controlled vocabularies, identifiers, units and document versions, and preserve original source values rather than normalising away a disagreement. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and field-by-field comparison. Decision interval: at booking acceptance, 24 hours before and two hours before arrival. Minimum evidence: unique booking and current values for stream, container, service method, vehicle, site instruction, access and contact route. Method: compare exact fields and versions, identify the source owner for each mismatch and produce a pre-arrival exception list. Uncertainty: show unit conversions, restricted fields, missing source versions and differences that may be harmless versus service-blocking. Safe failure: mark the booking not ready and refer each mismatch to its source owner; do not infer the intended scope.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The waste service manager owns operational scope; Contract Management owns contracted service definitions; the contractor owns vehicle and service details; the building manager owns site instructions; Security owns access requirements. Use current booking, contract schedule and effective site instruction. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Consistency expires on any changed field. Current site safety and access instructions outrank older booking notes; accepted contract variation outranks the base schedule. Unresolved mismatches block a ready state.

ANSWER BOUNDARY

Do not disclose commercial or personal contact details beyond role permission, amend booking or contract fields, or tell the contractor to proceed on conflicting instructions. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-054

STAKEHOLDER QUESTION

Were bins, skips, compactors, seals or locks exchanged and handed back exactly as recorded?

WHY THIS IS USEFUL

Supports asset control, safe follow-up and accurate charging after a contractor exchange or service visit.

SENSORS AND TELEMETRY

Modality and location: Paired pre- and post-service asset-tag scans, container controller identity where commissioned, seal or lock state and authorised handover events at the exchange point. Coverage, timing and aggregation: Cover every asset named in the service booking. Capture immediately before release and after hand-back or replacement; aggregate expected, removed, returned, substituted, seal-changed and unresolved states per asset. Measurement discipline: Verify tag serials, detect duplicate or reused identifiers, validate seal-state transitions and distinguish a controller communication loss from a substituted container. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: observation and asset-event reconciliation. Decision interval: immediately after each exchange or hand-back. Minimum evidence: booking asset list, pre-service state, post-service paired observations and authorised handover record. Method: compare expected and observed assets and state transitions, identify substitutions and generate an exception list tied to exact evidence. Uncertainty: show unobserved sides, missing scans, controller-health issues and whether a substitute is contractually allowed. Safe failure: leave the asset state unresolved, retain pre-service evidence and request an authorised physical and documentary confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The Estates waste lead owns the asset register; the contractor owns exchange and hand-back records; the site coordinator owns local handover; Contract Management owns allowed substitutions. Use current booking, asset register, signed handover and approved substitution rules. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Exchange evidence is current until the next asset event. A signed handover or authorised physical confirmation outranks a last-seen device state. Any unapproved substitution or seal discrepancy remains open until the owners resolve it.

ANSWER BOUNDARY

Do not accuse the contractor or staff of loss, open containers, alter seals or locks, accept an unauthorised substitute or update the asset register automatically. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

contractor coordination, storage and collections

QUESTIONS 05-06 OF 10

WM-055

STAKEHOLDER QUESTION

Will available storage and containment remain adequate until the next confirmed collection, including a plausible contractor delay?

WHY THIS IS USEFUL

Helps the team activate temporary authorised capacity or escalation before a store or container reaches its operational limit.

SENSORS AND TELEMETRY

Modality and location: Current fill or weight evidence, store capacity and segregation records, container seal or door state, environmental limits where specified and confirmed collection timing. Coverage, timing and aggregation: Cover all streams dependent on the next collection and the authorised storage locations. Refresh high-risk capacity at least hourly and model the confirmed window plus an owner-defined delay scenario; aggregate remaining capacity and time-to-limit ranges. Measurement discipline: Validate capacity against current usable volume, exclude unavailable sensors, confirm environmental and segregation constraints separately and do not treat nominal floor area as usable storage. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: forecast and multi-constraint scenario calculation. Decision interval: daily, after a collection change and at defined high-risk checkpoints. Minimum evidence: current usable capacity and limits, valid current fill or bounded estimate, confirmed collection window and a specified delay scenario. Method: project each stream to the next collection under central and conservative rates and test capacity, seal, segregation and environmental limits independently. Uncertainty: show demand range, delay assumption, sensor gaps, usable-capacity uncertainty and constraints not represented by volume. Safe failure: apply the current contingency, increase authorised checks and escalate to the storage owner without proposing unsafe overflow storage.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The Estates waste lead owns storage capacity; the competent specialist owner controls specialist limits; the contractor owns collection confirmation; Facilities and Fire Safety own current store restrictions. Use current store records, procedures, bookings and valid evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Adequacy expires on any deposit, collection, access, environmental or booking change. Safety and segregation limits outrank volume. If collection time is unconfirmed or a required limit is unknown, return not confirmed under the delay scenario.

ANSWER BOUNDARY

Do not designate an unapproved storage area, mix streams, extend specialist storage limits, guarantee capacity or book extra collection, labour or transport. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-056

STAKEHOLDER QUESTION

Which contractor tasks are incomplete, rejected or disputed, and what evidence supports the next operational follow-up?

WHY THIS IS USEFUL

Gives the team a precise exception list and evidence pack for corrective contact, without turning analytics into a contractual decision.

SENSORS AND TELEMETRY

Modality and location: Authorised service checklist events, asset or container scans, collection start and completion events, rejection codes, signed contractor notes and current site observations. Coverage, timing and aggregation: Cover each task in the relevant booking or work order. Capture evidence at the service event and aggregate completed, partial, rejected, disputed, not observed and not applicable states per task. Measurement discipline: Use the current task definition, require traceable event IDs and signatures, distinguish lack of evidence from incomplete work and preserve contractor and site statements separately. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup, comparison and evidence-pack preparation. Decision interval: after each service and before the next authorised contractor contact. Minimum evidence: current task list, original contractor evidence, authorised site observation and acceptance criteria for the named task. Method: compare expected and evidenced outcomes, separate operational fact from contractual interpretation and assemble the exact unresolved items. Uncertainty: show missing evidence, restricted commercial fields and whether the dispute concerns fact, scope, timing or acceptance. Safe failure: keep the task unresolved and refer the evidence pack to the authorised service or contract owner.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The waste service manager owns operational acceptance; Contract Management owns contractual interpretation; the contractor owns its service report; the site coordinator owns observations. Use current work order, contract schedule, signed service evidence and authorised exception records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Task status remains open until accepted or otherwise closed by the authorised owner. Contractual documents outrank dashboard interpretations. Conflicting site and contractor evidence is displayed side by side and cannot be resolved automatically.

ANSWER BOUNDARY

Do not accuse a named worker, determine liability, accept or reject a contractual claim, disclose commercial detail beyond permission or direct additional work. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

contractor coordination, storage and collections

QUESTIONS 07-08 OF 10

WM-057

STAKEHOLDER QUESTION

Which contracted services are repeatedly late, missed, incomplete or outside the booked scope, and what evidence supports formal contract action?

WHY THIS IS USEFUL

Provides a defensible pattern summary for the contract owner while separating operational evidence from commercial judgement.

SENSORS AND TELEMETRY

Modality and location: Matched booking, arrival, service-completion, task, rejection and exception events with contract-version identifiers; no driver or worker tracking. Coverage, timing and aggregation: Cover a defined contract period with stable service definitions, normally at least 12 scheduled events or three months. Aggregate performance by service type, stream and agreed metric, while retaining serious single exceptions separately. Measurement discipline: Validate event linkage, remove duplicates, split data at contract or service-definition changes and use agreed grace periods and exclusions exactly. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Contract Management or Procurement owns contract interpretation and action; the waste service manager owns operational records; the contractor owns its records; Finance owns credited or charged events. Use current contract schedules, accepted variations and original service evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Results apply only to the named contract version and period. Accepted variations and force-majeure or exclusion records are applied only when authoritative. Conflicting event evidence remains unresolved and is excluded or separately reported, not forced into a metric.

ANALYSIS REQUIRED

Operation: comparison and calculation. Decision interval: monthly, quarterly or at a contract-owner checkpoint. Minimum evidence: current contract definitions, at least 12 comparable scheduled events where feasible, traceable outcomes and accepted exclusions. Method: calculate agreed performance measures, confidence ranges and exception lists, then separate repeated pattern, isolated event and insufficient evidence. Uncertainty: show denominator, excluded events, record conflicts, contract changes and the difference between operational failure and contractual breach. Safe failure: state that the pattern is not established and provide the verified event list only.

ANSWER BOUNDARY

Do not determine breach, liability, penalty or supplier fitness, identify individual workers or initiate commercial action, withholding or termination. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-058

STAKEHOLDER QUESTION

Are the carrier, broker and receiving-facility records required for the planned service current and available to authorised staff?

WHY THIS IS USEFUL

Allows the authorised owner to identify an expired, missing or mismatched regulatory record before a collection proceeds.

SENSORS AND TELEMETRY

Modality and location: Authoritative registry or document-status lookup, contract record and booking linkage; no physical sensor can establish legal status. Coverage, timing and aggregation: Cover every carrier, broker and facility named in the planned service. Retrieve current records at booking and again before collection; aggregate identifier, scope, effective dates, source, retrieval time and match to the booking. Measurement discipline: Validate official identifiers and exact legal entity names, preserve source responses and treat inaccessible or ambiguous registry results as unavailable. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The regulatory body owns official records; Procurement or Contract Management owns supplier identity and contract scope; the contractor owns submitted documents; the waste compliance owner interprets applicability. Use current official sources and accepted internal copies only within permission. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Status is current only to the official source retrieval time and applicable scope. Official current records outrank supplier copies; a competent owner decides whether the record is required and sufficient. Mismatched names, scopes or dates prevent a ready answer.

ANALYSIS REQUIRED

Operation: lookup and identity-scope comparison. Decision interval: at onboarding, booking and immediately before the service. Minimum evidence: official identifier or exact entity identity, current official status, applicable service scope and current booking linkage. Method: match entities and scope, check effective dates and return current, expired, missing, mismatched, inaccessible or not determined. Uncertainty: show retrieval time, official-source availability, entity-matching uncertainty and scope that requires competent interpretation. Safe failure: return record status not confirmed and refer the planned service to the authorised compliance and contract owners.

ANSWER BOUNDARY

Do not certify legal compliance, decide that transport may proceed, disclose restricted supplier records broadly or substitute a web or supplier claim for the competent owner. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

contractor coordination, storage and collections

QUESTIONS 09-10 OF 10

WM-059

STAKEHOLDER QUESTION

Which transfer notes, consignment records, weight tickets or destruction certificates are due, received, matched or unresolved for recent collections?

WHY THIS IS USEFUL

Supports timely document chasing and prevents a collection from being closed or reported with a missing or mismatched evidence item.

SENSORS AND TELEMETRY

Modality and location: Document-receipt events, booking and consignment identifiers, authorised electronic signatures and source-system timestamps; no waste-content sensing. Coverage, timing and aggregation: Cover every collection in the current retention and reporting window. Capture issue, receipt, supersession and acceptance events; aggregate due date, received state, identifier match, version and unresolved field differences per document. Measurement discipline: Validate document identity, signatures, date order, stream and weight fields, detect duplicates and retain original files and supersession relationships. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and document reconciliation. Decision interval: daily for open collections and before reporting, payment or closure. Minimum evidence: traceable collection, current required-document list and original received records with identifiers and dates. Method: match documents to collection and stream, check required fields and return due, received-unchecked, matched, mismatched, superseded or unresolved. Uncertainty: show missing fields, extraction uncertainty, duplicate versions and any restricted document content. Safe failure: keep the collection open in the evidence register and request the issuing or accepting owner to resolve the item.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The waste compliance owner owns required-document rules; the contractor or carrier owns issued documents; the service manager owns collection events; Finance owns invoice linkage; Records Management owns retention. Use original signed records and current requirement matrices. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A document is current until superseded or rejected by the authorised owner. Original signed documents outrank extracted dashboard fields. Conflicting versions remain linked and unresolved until the owner accepts one; do not silently overwrite.

ANSWER BOUNDARY

Do not forge, edit, accept or reject a document, claim destruction from a booking or invoice, disclose confidential records or certify legal completeness. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-060

STAKEHOLDER QUESTION

What must the next shift know about pending collections, storage pressure, failed assets, access constraints and unresolved escalations?

WHY THIS IS USEFUL

Creates a concise operational handover so urgent exceptions are not lost between teams or service periods.

SENSORS AND TELEMETRY

Modality and location: Current exception records, container and store status, confirmed bookings, route and access state, asset-health alerts and authorised owner actions; no person-level performance or confidential-content detail. Coverage, timing and aggregation: Cover all open items that may affect the next shift. Refresh at handover and on critical events; aggregate by urgency, required action owner, due time, evidence age and dependency. Measurement discipline: Deduplicate linked alerts, verify that closed items are excluded, align event times and retain the original source link for each handover statement. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup, prioritisation and summary generation. Decision interval: at each shift or service-period handover and after a critical event. Minimum evidence: current open source records, accountable owner, due time, evidence age and the receiving role's permission. Method: group items into act now, monitor, await owner and information gap, then order by current consequence and time dependency. Uncertainty: show snapshot time, stale inputs, unresolved conflicts and withheld restricted detail. Safe failure: issue an incomplete handover flag, preserve direct source links and require verbal or owner confirmation for critical unknowns.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Each operational source owner retains authority for its item; the waste service manager owns the handover compilation. Use current open exceptions, bookings, storage status, fault records and competent directions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The handover snapshot expires at the stated cut-off and critical events must update it. Source records outrank the summary. Conflicting or stale items are labelled and retained rather than rewritten as a single fact.

ANSWER BOUNDARY

Do not identify or rank staff, disclose confidential waste or access details beyond need, invent a closure or promise that an action will be completed. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

weights, costs, provenance and reporting

WM-061

STAKEHOLDER QUESTION

What measured or estimated weight was collected for each waste stream and reporting period?

WHY THIS IS USEFUL

Provides a stream-by-period figure that clearly separates direct weighing from allocation or estimation.

SENSORS AND TELEMETRY

Modality and location: Calibrated weighbridge or platform-scale records, on-vehicle weighing where contractually supported, container load cells where validated and collection-event identifiers; no conversion from fill level to mass unless a supported density model is stated. Coverage, timing and aggregation: Cover every collection in the selected period and stream scope. Capture original gross, tare or net values and collection events; aggregate direct weight, allocated weight, estimated weight and unavailable weight separately. Measurement discipline: Check calibration validity at the weighing time, units, tare method, zero checks, duplicate tickets and clock sequence; validate any density conversion against local paired weights. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The scale operator or contractor owns original weight tickets; the Estates waste lead owns stream and site allocation; Finance owns invoice-linked figures; Sustainability owns reporting definitions. Use original weighing records, collection events and current allocation rules. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A weight is current to the original event and can be superseded only by an authorised corrected record. Direct valid net weight outranks allocation, which outranks model estimate. Conflicting tickets remain visible and the collection is unresolved until the source owner confirms.

ANALYSIS REQUIRED

Operation: lookup, calculation and estimate. Decision interval: for each collection and each defined reporting close. Minimum evidence: traceable collection, stream, valid direct weight or a current approved allocation or estimation rule, and complete units and time fields. Method: calculate net weight where gross and tare are valid, otherwise apply the named allocation or model and label the result by evidence class. Uncertainty: show scale tolerance, allocation share, model error, missing collections and confidence at stream and period level. Safe failure: report weight unavailable for unsupported collections and do not substitute nominal container capacity.

ANSWER BOUNDARY

Do not present estimated weight as measured, infer contents from mass, edit source tickets, certify official tonnage or release a figure without the reporting owner's permission. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-062

STAKEHOLDER QUESTION

How should shared vehicle or campus-level weights be allocated to Abacws without false precision?

WHY THIS IS USEFUL

Enables transparent local reporting when direct building-level weighing does not exist and prevents an arbitrary share being presented as fact.

SENSORS AND TELEMETRY

Modality and location: Shared net-weight records, traceable collection stops, container counts or validated pre-collection weights, route and stream identifiers and approved building-service factors; no person-level or room-level inference. Coverage, timing and aggregation: Cover each shared collection that includes Abacws and all contributing sites required by the chosen rule. Capture original total weight and valid allocation inputs; aggregate allocated share, method and residual by stream and period. Measurement discipline: Validate stop linkage, units and completeness, use only inputs measured or recorded in the same collection window and test whether the allocation shares sum to the original total. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The contractor owns shared weight and route evidence; the Estates waste lead owns Abacws container and service data; Sustainability and Finance own approved allocation policy. Use the current documented rule and original source records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Allocation is valid only for the named route, stream and period. Direct building weight outranks allocation. If input coverage is incomplete or methods conflict, produce a range or leave the share unavailable; never select the most favourable method silently.

ANALYSIS REQUIRED

Operation: calculation, estimate and scenario comparison. Decision interval: per shared collection and at reporting close. Minimum evidence: valid shared total, traceable Abacws participation and at least one approved allocation basis with coverage for all required contributors. Method: apply the approved rule, reconcile shares to the source total and compare plausible alternative rules where the owner has not fixed one. Uncertainty: show method, share basis, residual, sensitivity range and any contributor with missing data. Safe failure: report no defensible Abacws allocation and retain the shared total only at its authoritative scope.

ANSWER BOUNDARY

Do not invent a building share, imply room-level precision, use occupancy as a proxy without owner approval or certify the allocation as a measured building weight. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

weights, costs, provenance and reporting

QUESTIONS 03-04 OF 10

WM-063

STAKEHOLDER QUESTION

Which scale or weighing method is valid for each reported weight, and is its calibration and traceability evidence sufficient for operational use?

WHY THIS IS USEFUL

Stops expired, misapplied or weak weighing evidence from entering cost and sustainability reports.

SENSORS AND TELEMETRY

Modality and location: Scale identity, calibration certificate or check result, zero and reference-load tests, weighing event metadata, tare method and device-health records at the weighbridge, vehicle, platform or container. Coverage, timing and aggregation: Cover every weighing device and method contributing to the selected report. Evaluate status at each weighing time and aggregate valid, restricted, expired, failed or unknown evidence state per event and device. Measurement discipline: Check calibration scope, validity dates, units, capacity, resolution, environmental limits, zero and reference tests, clock offset and whether the device was used within its approved range. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and metrological validity calculation. Decision interval: for every reported weighing event and before period release. Minimum evidence: unique device and event, original calibration or check evidence, tare method, operating range and acceptance rule for the intended use. Method: test identity, date, scope, range, checks and traceability and assign valid, restricted, invalid or unknown to each weight. Uncertainty: show measurement uncertainty, certificate scope, missing checks and whether the status affects operational, financial or external reporting use. Safe failure: exclude the weight from the named calculation or retain it only as an explicitly unsupported observation.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The scale owner or contractor owns calibration and operating records; the waste reporting owner defines accepted methods; Finance owns payment-use requirements; suppliers provide specifications. Use original certificates, check records and weighing events. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Validity is evaluated at the event time, not only at retrieval. A competent restriction or failed check outranks an unexpired certificate. If device identity, tare method or certificate scope cannot be matched, the weight is not valid for the named use.

ANSWER BOUNDARY

Do not alter calibration records, declare legal metrology compliance, assume a certificate covers another device or use, or convert an invalid weight into a precise estimate. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-064

STAKEHOLDER QUESTION

What is the evidenced cost per lift, collection, container and tonne for each stream, including known credits and surcharges?

WHY THIS IS USEFUL

Lets the team identify the cost structure behind a stream and prepare a transparent option for budget and contract owners.

SENSORS AND TELEMETRY

Modality and location: No physical sensor establishes cost. Use matched collection events, valid weights, container service records and authorised invoice, credit and surcharge lines with exact contract codes. Coverage, timing and aggregation: Cover the selected contract, stream and reporting period. Capture original financial and service events and aggregate gross cost, credits, surcharges, net cost and unit measures by evidence class. Measurement discipline: Validate currency, tax treatment, units, contract version, duplicate invoice lines and linkage between each charge and the related collection, container or weight. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

ANALYSIS REQUIRED

Operation: lookup and deterministic calculation. Decision interval: monthly and before budget or contract-owner decisions. Minimum evidence: authorised financial records, current contract pricing, matched service events and valid denominators for the selected unit measure. Method: reconcile charges and credits, calculate unit costs only where numerator and denominator share scope and period, and label unmatched or estimated components. Uncertainty: show missing weights, allocation effects, tax and credit treatment, disputed lines and the impact of small denominators. Safe failure: report the unit cost unavailable or as a bounded provisional range and refer unresolved lines to Finance or Contract Management.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Finance owns posted financial records; Contract Management owns price schedules and variations; the waste service manager owns service events; the contractor owns invoices and credits. Use current ledger extracts, accepted contract schedules and traceable operational evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Costs are current to the posted or accepted record and named contract version. Posted ledger entries outrank estimates; accepted variations outrank base prices. Disputed or unmatched lines remain separate and are not included in an asserted unit cost.

ANSWER BOUNDARY

Do not expose restricted commercial terms beyond permission, approve payment, determine value for money, promise savings or present an unmatched invoice line as a service cost. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

weights, costs, provenance and reporting

WM-065

STAKEHOLDER QUESTION

Which invoices, credits, weight records and collection events disagree, and what is the exact mismatch?

WHY THIS IS USEFUL

Creates a precise reconciliation queue for Finance, the service owner and the contractor instead of a vague total difference.

SENSORS AND TELEMETRY

Modality and location: Original invoice and credit lines, valid weight tickets, booking and completion events, container identifiers and accepted contract price codes; no physical sensing beyond source service and weighing evidence. Coverage, timing and aggregation: Cover all financial lines and service events in the selected reconciliation window. Recalculate after any source correction and aggregate unmatched, duplicate, quantity, price, stream, date and identifier mismatches. Measurement discipline: Normalise units and dates without overwriting originals, test duplicate identifiers, preserve credit links and require traceable many-to-one or one-to-many mappings where legitimate. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Finance owns ledger and payment status; Contract Management owns price interpretation; the waste service manager owns collection events; the contractor owns invoice and weight evidence. Use original source records and accepted contract variations. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: A mismatch remains open until an authorised source correction or acceptance event. Posted ledger and signed operational records retain their original states; neither is silently altered to force agreement. Conflicting corrections are versioned and exposed.

ANALYSIS REQUIRED

Operation: comparison and reconciliation. Decision interval: weekly for open items and at each accounting close. Minimum evidence: traceable financial line, related service or weight evidence, current price schedule and unique identifiers or an approved aggregation rule. Method: match records, compute expected versus charged quantities and values, classify the mismatch and identify the responsible source owner. Uncertainty: show tolerance, unit conversion, allocation, timing difference and any ambiguous many-to-one link. Safe failure: leave the item unresolved, exclude it from asserted reconciled totals and route the evidence to the relevant owners.

ANSWER BOUNDARY

Do not edit source records, approve or withhold payment, assign fault, disclose commercial details broadly or claim fraud or overcharging from a mismatch alone. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-066

STAKEHOLDER QUESTION

Which waste records are missing, duplicated, late, reversed, misdated or linked to the wrong stream, container or collection?

WHY THIS IS USEFUL

Provides a data-quality work queue so operational and reporting figures are corrected at source rather than patched downstream.

SENSORS AND TELEMETRY

Modality and location: Source-system event identifiers, timestamps, version and supersession links, stream and container mappings, gateway receipt times and document metadata; no new physical sensing is required. Coverage, timing and aggregation: Cover all records feeding the selected operational or reporting period. Run checks on ingestion and daily; aggregate exceptions by source, field, collection and consequence. Measurement discipline: Test uniqueness, event order, allowed lateness, mapping validity, reversal logic, clock offset and required-field completeness while preserving original values. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Each source-system owner owns its records; the Estates waste lead owns common asset and stream mappings; the reporting owner owns quality rules. Use original source events, current mapping versions and authorised correction records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Quality state changes only on a new source event or authorised correction. Original records and formal supersession links outrank transformed copies. Conflicting corrections remain unresolved and must not be collapsed.

ANALYSIS REQUIRED

Operation: diagnosis of record quality. Decision interval: on ingestion, daily and before any report release. Minimum evidence: original record IDs, current mapping and timing rules, source versions and complete transformation lineage. Method: apply deterministic quality tests, classify each exception and calculate its affected outputs and source owner. Uncertainty: show false-positive risk, clock tolerance, source outages, incomplete lineage and whether the issue affects count, weight, cost or status. Safe failure: exclude affected fields or records from calculations, preserve the exception and request correction at the authoritative source.

ANSWER BOUNDARY

Do not fabricate missing values, overwrite originals, infer the intended stream or container without evidence, or blame a person or supplier for a data-quality exception. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

QUESTIONS 07-08 OF 10

What source records, assumptions, allocation rules and transformations produced each dashboard or report figure?

Lets an authorised reader reproduce a figure, locate weak assumptions and decide whether it is suitable for the intended decision.

Modality and location: No additional sensing is required. Use source-record identifiers, dataset versions, transformation code or rule versions, allocation parameters, run timestamps and permission-aware lineage links. Coverage, timing and aggregation: Cover every displayed figure and its complete calculation path. Capture lineage at each run and aggregate source counts, excluded records, assumptions, transformation steps and output hash per figure. Measurement discipline: Verify that lineage points to immutable or versioned sources, test output hashes, record software and rule versions and identify any manual value or spreadsheet step explicitly. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Trace evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

Owners and sources: Source owners retain authority for inputs; the reporting owner owns definitions and transformations; Information Governance owns access; Finance or Sustainability owns release for their uses. Use original records, versioned rules and authorised run artefacts. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Lineage is valid only for the exact figure version and run time. Original source records outrank copied extracts; an authorised definition change creates a new figure version. Missing lineage makes the figure non-reproducible.

Operation: lookup and reproducibility check. Decision interval: for every generated figure and before sharing or sign-off. Minimum evidence: figure ID and version, source IDs and versions, transformation and allocation rules, exclusions, run time and output hash. Method: trace every output field to source records, replay deterministic steps where permitted and report complete, partial or missing lineage. Uncertainty: show restricted source links, non-deterministic steps, manual interventions and unavailable historical versions. Safe failure: label the figure non-reproducible or partially reproducible and prevent it from being presented as fully traceable.

Do not disclose restricted source data, invent lineage, alter historical inputs, or claim that reproducibility proves correctness, compliance or suitability. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

What recycling, preparation-for-reuse, recovery and disposal figures can be reported for the selected period, and with what confidence?

Provides an honest destination and treatment summary that separates supported outcomes from assumptions and missing evidence.

Modality and location: Validate stream weights or allocations, contractor destination and treatment codes, reuse handover records and receiving evidence; no sensor can prove final treatment on its own. Coverage, timing and aggregation: Cover every in-scope collection and reuse movement in the selected period. Aggregate mass by treatment category and evidence class, while retaining unknown destination or treatment separately. Measurement discipline: Validate code mappings, prevent double counting across transfer stages, reconcile totals to source weight and require receiving or contractor evidence for final-treatment claims. Use traceable calibration for measured quantities or a versioned functional check for discrete events. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

Owners and sources: Sustainability owns reporting definitions; the contractor or receiving facility owns treatment evidence; the Estates waste lead owns stream and site scope; Finance owns relevant credits; Records Management owns retained documents. Use current definitions and traceable original evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Figures are valid only for the stated period, scope and destination evidence. Direct receiving evidence outranks contractor estimate, which outranks generic stream assumptions. Conflicting or missing treatment evidence remains unknown and is not assigned to a favourable category.

Operation: calculation and estimate. Decision interval: monthly, quarterly and at authorised reporting close. Minimum evidence: complete in-scope collection list, valid weights or allocations, current treatment mapping and evidence class for each movement. Method: reconcile total mass, assign treatment only where supported, calculate rates with unknown mass visible and provide confidence by category. Uncertainty: show allocation error, missing weights, treatment-code uncertainty, double-count prevention and sensitivity to unknown mass. Safe failure: report only supported mass and an explicit unknown category; do not calculate a definitive rate from incomplete destination evidence.

Do not certify recycling or recovery, assume a bin label equals final treatment, hide unknown mass or release figures beyond the authorised reporting scope. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

weights, costs, provenance and reporting

QUESTIONS 09-10 OF 10

WM-069

STAKEHOLDER QUESTION

Which reporting periods or service points are not directly comparable because contracts, streams, occupancy context, measurement methods or service definitions changed?

WHY THIS IS USEFUL

Prevents misleading trend claims and tells the team where a like-for-like comparison or explicit adjustment is possible.

SENSORS AND TELEMETRY

Modality and location: Versioned contract, stream, asset, route, measurement and operating-context records linked to each observation and reporting period; no additional physical sensing is required. Coverage, timing and aggregation: Cover every period or point in the proposed comparison. Capture effective dates and versions and aggregate comparability flags for scope, definition, method, capacity, context and data coverage. Measurement discipline: Validate change dates, distinguish gradual data drift from formal definition changes and retain original units and classifications before harmonisation. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Contract Management owns contract definitions; Sustainability owns stream and reporting definitions; the Estates waste lead owns assets and routes; source-system owners own methods; building and timetable owners supply context. Use current and historical versioned records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Comparability is assessed for the named metric and question, not globally. Formal effective records outrank inferred change points. Where change dates conflict, split the period conservatively and mark the comparison unresolved.

ANALYSIS REQUIRED

Operation: lookup and comparability assessment. Decision interval: before every period, site or station comparison. Minimum evidence: metric definition, scopes, source methods, effective versions and coverage for every compared unit. Method: test a comparability matrix, identify exact breaks and either allow direct comparison, require a stated adjustment or reject comparison. Uncertainty: show residual differences, adjustment assumptions, missing version history and whether only direction rather than magnitude is comparable. Safe failure: do not calculate a combined trend; present separate period figures with the comparability gaps.

ANSWER BOUNDARY

Do not imply that unlike periods are equivalent, back-cast unsupported definitions, use the May 2021 plan as current operating context or claim a trend from incomparable data. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-070

STAKEHOLDER QUESTION

What data correction, source-owner confirmation or formal sign-off is still needed before a waste figure is released or submitted?

WHY THIS IS USEFUL

Gives the reporting owner a clear release gate and prevents unresolved evidence from being presented as final.

SENSORS AND TELEMETRY

Modality and location: Current data-quality exceptions, unresolved source conflicts, lineage status, permission state and required owner sign-off events; no new physical sensing is required unless a named gap calls for observation. Coverage, timing and aggregation: Cover every figure in the proposed release. Recalculate on each source or sign-off event and aggregate blocking, non-blocking and informational conditions by figure and release purpose. Measurement discipline: Use versioned release criteria, verify signatory roles and expiry, ensure the figure hash matches the assessed version and preserve unresolved source evidence. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The reporting owner controls release; source owners control corrections; Sustainability, Finance, Information Governance or other named owners provide purpose-specific sign-off. Use current release criteria, source records, lineage and authorised approvals. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Release status expires if the figure, source, definition or permission changes. Required competent or accountable sign-off outranks an analytic ready score. Any blocking conflict, missing lineage or expired permission prevents a ready answer.

ANALYSIS REQUIRED

Operation: lookup and release-gate calculation. Decision interval: before each internal or external release and after any material change. Minimum evidence: exact figure version, purpose, current release criteria, resolved quality status, complete lineage, permissions and required sign-offs. Method: evaluate each gate explicitly, list the responsible owner and return ready, blocked, restricted or unknown. Uncertainty: show which issues affect accuracy, traceability, permission or interpretation and whether they change the figure value. Safe failure: hold the figure, preserve the assessed version and request the named correction or sign-off.

ANSWER BOUNDARY

Do not sign on behalf of an owner, release restricted figures, certify compliance, suppress an unresolved conflict or describe a blocked figure as final. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

prevention, reuse, recycling and emissions reduction

QUESTIONS 01-02 OF 10

WM-071

STAKEHOLDER QUESTION

After a verified prevention or reuse change, where could bin capacity or collection frequency be reduced through a controlled service test without increasing overflow, hygiene or access risk?

WHY THIS IS USEFUL

Connects waste reduction to a cautious operational adjustment with explicit safeguards and a reversible decision.

SENSORS AND TELEMETRY

Modality and location: Commissioned fill or weight evidence, verified service resets, overflow and hygiene events, asset and route records and the dated authorised prevention or reuse intervention; no individual behaviour sensing. Coverage, timing and aggregation: Cover candidate and comparison points for at least six valid cycles before and six after the intervention, then during the service test. Aggregate fill-at-service, threshold exposure, service effort, hygiene events and access issues by cycle. Measurement discipline: Verify intervention and service-change dates, maintain stable measurement definitions, recheck sensor geometry after capacity changes and exclude untraceable cycles. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The prevention or reuse owner owns the intervention; the waste service manager owns service-test limits; the Estates waste lead owns capacity; Hygiene, Accessibility and Fire Safety owners control constraints. Use authorised test plan, current service data and verified outcomes. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The test result applies only to the named intervention, station and period. Any safety, hygiene, access or overflow trigger ends the recommendation under the pre-stated stop rule. Confounded periods remain inconclusive.

ANALYSIS REQUIRED

Operation: recommendation, scenario calculation and controlled comparison. Decision interval: at test design, each checkpoint and test close. Minimum evidence: verified intervention, stable baseline, current constraints, pre-stated target and stop rules, and valid outcome evidence. Method: model a conservative service option, obtain owner authorisation outside the service, then compare outcome ranges and stop-rule events during the test. Uncertainty: show seasonal effects, intervention uptake uncertainty, sensor gaps and that a service change may not be attributable to the intervention alone. Safe failure: retain the existing authorised capacity and frequency or revert under the stop rule; state that reduction is not supported.

ANSWER BOUNDARY

Do not change service automatically, promise savings or emissions benefits, infer user behaviour or continue a test after a competent safety, hygiene or access stop. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-072

STAKEHOLDER QUESTION

Which disposable products or waste streams show the strongest evidence-backed opportunity for prevention at source?

WHY THIS IS USEFUL

Helps the team focus procurement and service-owner discussion on high-volume, high-cost or avoidable streams with traceable evidence.

SENSORS AND TELEMETRY

Modality and location: Validated stream weights or allocations, purchase and issue records at an appropriate aggregate level, service counts, contractor treatment evidence and coded waste-composition studies only where separately authorised and privacy-safe. Coverage, timing and aggregation: Cover a defined period, product group and service boundary. Aggregate purchased or issued quantity, discarded mass, collection cost and avoidability evidence by product or stream, without linking to individuals or inspecting confidential contents. Measurement discipline: Match units and periods, separate purchased from used and discarded quantities, treat composition samples as bounded estimates and exclude categories that cannot be observed safely. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Procurement owns purchase data; service or product owners own issue and use policy; Sustainability owns prevention definitions; the Estates waste lead owns waste evidence; Finance owns cost. Use current source records and approved category mappings. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Opportunity ranking is valid only for the stated scope and data coverage. Direct product-specific evidence outranks generic sector assumptions. Missing use or composition evidence widens the range and may limit the result to a stream-level opportunity.

ANALYSIS REQUIRED

Operation: comparison and opportunity estimate. Decision interval: quarterly and before procurement or service-design decisions. Minimum evidence: at least one complete representative period, traceable waste and purchase or issue evidence, current costs and an authorised prevention lever. Method: calculate mass, cost and frequency ranges, assess practical avoidability and rank opportunities using transparent criteria rather than a hidden score. Uncertainty: show allocation error, stock changes, unobserved disposal routes, category-mapping uncertainty and non-comparable periods. Safe failure: report only broad stream opportunities or no supported priority and request targeted aggregate evidence.

ANSWER BOUNDARY

Do not inspect confidential waste, identify users or departments from discarded items, dictate purchasing, claim all discarded product was avoidable or authorise a product change. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

prevention, reuse, recycling and emissions reduction

QUESTIONS 03-04 OF 10

WM-073

STAKEHOLDER QUESTION

Which discarded furniture, IT, equipment or consumables may enter an authorised reuse, refurbishment, donation or take-back route?

WHY THIS IS USEFUL

Prevents potentially reusable assets from entering waste while keeping ownership, safety, security and value decisions with the authorised teams.

SENSORS AND TELEMETRY

Modality and location: Asset tags, authorised discard requests, external condition codes, dimensions, seal or custody status and approved reuse-system records; no access to device data, confidential labels or item history beyond role permission. Coverage, timing and aggregation: Cover each item or batch presented for disposal. Capture at discard request, secure holding, owner decision and handover; aggregate eligibility conditions, missing owner actions and authorised route options per item or batch. Measurement discipline: Verify identity and ownership, use versioned external condition codes and do not infer electrical safety, data sanitisation or fitness from appearance. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The asset owner controls release; Procurement or Finance controls ownership and value rules; Information Security controls data-bearing items; Health and Safety or technical owners control fitness; Sustainability owns reuse routes. Use current asset, release and route records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Eligibility expires after condition, custody, ownership or route change. Security, safety and asset-owner restrictions outrank reuse preference. Any missing release or competent decision results in held, not eligible for automatic routing.

ANALYSIS REQUIRED

Operation: lookup and eligibility-gate calculation. Decision interval: at discard request, before movement and before reuse handover. Minimum evidence: unique item or batch, current owner release, applicable safety and security state, external condition and an active authorised route. Method: test each eligibility condition and return candidate, held, not eligible or unknown with the responsible owner for each gap. Uncertainty: show unassessed internal condition, restricted fields, batch assumptions and route capacity or acceptance uncertainty. Safe failure: keep the item in its current authorised holding state and request the missing owner or competent decision.

ANSWER BOUNDARY

Do not access data, test or declare equipment safe, transfer ownership, value the item, remove asset labels or authorise reuse, donation, take-back or disposal. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-074

STAKEHOLDER QUESTION

Which approved reused items are available now to meet an internal need before a new purchase is requested?

WHY THIS IS USEFUL

Allows an authorised requester to consider a verified reuse option without relying on stale stock or unsuitable items.

SENSORS AND TELEMETRY

Modality and location: Reuse inventory status, asset or batch tags, current storage location at permitted granularity, reservation state and authorised external condition and specification records; no person tracking or data access. Coverage, timing and aggregation: Cover items in the approved reuse inventory and the stated need. Refresh availability on every reservation, movement or handover and aggregate matching specification, quantity, condition, location zone and reservation state. Measurement discipline: Verify tag-to-record identity, split reserved from available stock, check specification units and treat missing condition or current location as unavailable. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The reuse service owner owns inventory; the asset owner controls release; Procurement owns need and purchasing workflow; technical, safety and information-security owners control suitability. Use current reuse inventory, authorised need specification and owner decisions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Availability expires on any reservation, movement or condition change. Competent suitability and security restrictions outrank specification matching. If location or custody is uncertain, the item is not presented as available.

ANALYSIS REQUIRED

Operation: lookup and constrained matching recommendation. Decision interval: at the point of an authorised internal need and before reservation. Minimum evidence: current need specification, available authorised inventory, valid condition and suitability state and role permission. Method: match mandatory specifications first, then present eligible options with differences and next owner action; do not auto-reserve. Uncertainty: show stock age, unverified attributes, condition limits, location granularity and whether suitability needs competent confirmation. Safe failure: return no confirmed reuse match and allow the normal procurement workflow to continue.

ANSWER BOUNDARY

Do not reserve, allocate or transfer an item, claim equivalent performance, disclose restricted asset locations or pressure a requester to accept a reuse option. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

prevention, reuse, recycling and emissions reduction

QUESTIONS 05-06 OF 10

WM-075

STAKEHOLDER QUESTION

Did a prevention, reuse or refurbishment intervention reduce purchasing and waste after accounting for seasonality and service changes?

WHY THIS IS USEFUL

Provides a cautious outcome estimate for deciding whether to continue, adapt or stop an intervention.

SENSORS AND TELEMETRY

Modality and location: Versioned intervention records, aggregate purchase or issue quantities, validated waste weights or allocations, reuse handovers, service changes and approved operating-context labels; no individual monitoring. Coverage, timing and aggregation: Cover a pre-stated baseline and follow-up period with at least three comparable reporting units on each side, preferably with a comparison group. Aggregate purchase, waste, reuse and service measures using unchanged definitions. Measurement discipline: Verify intervention timing, split periods at contract or measurement changes, reconcile units and retain missing or reclassified streams rather than smoothing them. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The intervention owner owns scope and timing; Procurement owns purchase evidence; the Estates waste lead owns waste records; Sustainability owns outcome definitions; Finance owns cost. Use current versioned records and authorised comparator definitions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Results apply only to the named intervention and context. Formal service or definition changes must be accounted for explicitly. Without a credible comparator or stable trend, report association or observed change, not causal effect.

ANALYSIS REQUIRED

Operation: comparison and impact estimate. Decision interval: at pre-stated checkpoints and intervention close. Minimum evidence: authorised intervention date and scope, comparable baseline and follow-up evidence, stable definitions or explicit adjustments and known service changes. Method: calculate absolute and relative ranges, use interrupted trend or matched comparison where supported and test sensitivity to seasonality and missing data. Uncertainty: show confidence interval, comparator weakness, concurrent changes, allocation error and whether evidence supports causation, association or description only. Safe failure: report the result as inconclusive or descriptive and avoid recommending scale-up solely from the observed difference.

ANSWER BOUNDARY

Do not claim personal behaviour change, guarantee future savings or emissions reductions, suppress negative outcomes or authorise continuation, expansion or closure. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-076

STAKEHOLDER QUESTION

Which unopened, obsolete or repeatedly unused consumables could be prevented through ordering, stock rotation or authorised redistribution?

WHY THIS IS USEFUL

Helps product and service owners reduce avoidable disposal while protecting stock quality, ownership and operational need.

SENSORS AND TELEMETRY

Modality and location: Aggregate stock and issue records, batch or product identifiers, expiry or support dates, authorised condition and storage-state records and waste-disposal events; no inspection of personal work areas or attribution to individuals. Coverage, timing and aggregation: Cover the selected consumable categories and authorised stores over at least one complete ordering cycle. Aggregate stock age, issue rate, expiry risk, disposal quantity and redistribution eligibility by product and store. Measurement discipline: Validate units, batch identifiers and stock snapshots, distinguish slow-moving from obsolete and treat missing issue records as unknown use rather than unused stock. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The stock or service owner controls operational need and redistribution; Procurement owns ordering records; Finance owns value; Sustainability owns prevention framing; Health and Safety or technical owners control storage and expiry. Use current inventory and disposal records. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Opportunity status expires on stock issue, condition, expiry or policy change. Safety, quality and service-need restrictions outrank waste prevention. Conflicting stock and issue records reduce the result to a reconciliation need.

ANALYSIS REQUIRED

Operation: comparison, forecast and recommendation. Decision interval: monthly and before the next ordering cycle. Minimum evidence: current stock, at least one ordering and issue cycle, expiry or obsolescence criteria, disposal evidence and an authorised prevention route. Method: forecast excess under low, central and high issue rates, identify stock rotation or redistribution options and state the owner action required. Uncertainty: show demand variability, unrecorded issues, batch condition, value basis and whether expiry or obsolescence is formal or inferred. Safe failure: make no prevention recommendation; request stock reconciliation or owner confirmation and retain current ordering controls.

ANSWER BOUNDARY

Do not access personal storage, accuse a team of over-ordering, redistribute or write off stock, declare product quality or safety, or change procurement settings. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

prevention, reuse, recycling and emissions reduction

QUESTIONS 07-08 OF 10

WM-077

STAKEHOLDER QUESTION

Which current supplier, producer-responsibility or contractor take-back route applies to a specific purchased item or material?

WHY THIS IS USEFUL

Gives the authorised owner a traceable route option before the item enters a general waste stream.

SENSORS AND TELEMETRY

Modality and location: Product and purchase identifiers, supplier or manufacturer take-back terms, contract records, item condition and authorised release status; no sensing can establish eligibility by itself. Coverage, timing and aggregation: Cover the named item or material and current supplier or producer schemes. Retrieve terms at the decision time and aggregate scheme scope, dates, condition, logistics, cost and evidence requirements. Measurement discipline: Match exact product, supplier, legal entity and purchase channel, preserve original terms and treat marketing statements without binding scope as informational only. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Procurement owns purchase and supplier identity; the supplier or producer owns scheme terms; Contract Management owns contractual routes; the asset or service owner controls release; Sustainability owns hierarchy reporting. Use current official terms and accepted contracts. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: Eligibility is current only to the retrieved terms and item state. Contractual terms and current scheme rules outrank generic manufacturer pages. If product identity, purchase route or scheme scope is uncertain, return no confirmed route.

ANALYSIS REQUIRED

Operation: lookup and eligibility comparison. Decision interval: before disposal booking or replacement purchase. Minimum evidence: exact item or material identity, current owner release, purchase or supplier linkage and effective take-back terms. Method: match eligibility conditions, required evidence, logistics and restrictions and present confirmed routes with owner actions. Uncertainty: show retrieval time, scheme-change risk, identity uncertainty, potential charges and conditions needing supplier confirmation. Safe failure: return no verified take-back route and use the current authorised waste route pending Procurement or supplier confirmation.

ANSWER BOUNDARY

Do not promise supplier acceptance, disclose restricted contract terms, book transport, transfer ownership or represent a take-back route as automatically superior or compliant. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-078

STAKEHOLDER QUESTION

Which waste stream could move higher in the waste hierarchy through a practical change without increasing contamination, safety or service risk?

WHY THIS IS USEFUL

Supports a balanced change proposal that considers prevention, reuse and recycling alongside operational constraints.

SENSORS AND TELEMETRY

Modality and location: Validated stream weights or allocations, contamination and rejection evidence, current treatment routes, service cost, collection capacity and authorised safety, storage and access constraints. Coverage, timing and aggregation: Cover a defined stream and representative period plus at least one feasible higher-hierarchy option. Aggregate mass, treatment, contamination, service burden, cost and constraint evidence by option. Measurement discipline: Reconcile mass and destination, ensure options use matched scope and period, separate observed outcomes from supplier claims and do not use generic emission or recycling factors as proof of local feasibility. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability owns hierarchy definitions and objective; the Estates waste lead owns operations; Health and Safety and specialist owners control safety; Procurement and Contract Management own supplier options and cost. Use current routes, contracts and operational evidence. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The recommendation is valid only for the named stream, option and current constraints. Safety and competent restrictions outrank hierarchy preference. Missing destination, acceptance or contamination evidence can prevent an option being supported.

ANALYSIS REQUIRED

Operation: comparison and multi-criteria recommendation. Decision interval: during annual planning or before a contract, product or service change. Minimum evidence: representative baseline, current treatment evidence, at least one feasible option, current costs and all material safety, access and contamination constraints. Method: compare options using transparent mass, service, risk and cost criteria, test sensitivity and identify a bounded next test or owner decision. Uncertainty: show unknown destination, supplier-performance uncertainty, contamination sensitivity and trade-offs between hierarchy, cost and service. Safe failure: make no hierarchy-change recommendation and preserve the current authorised route while gathering the missing feasibility evidence.

ANSWER BOUNDARY

Do not declare a route environmentally superior from a label alone, weaken safety or confidentiality controls, promise acceptance or savings, or authorise a stream or contract change. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

prevention, reuse, recycling and emissions reduction

QUESTIONS 09-10 OF 10

WM-079

STAKEHOLDER QUESTION

What defensible greenhouse-gas emissions range is associated with waste collection and treatment for the selected scope and period?

WHY THIS IS USEFUL

Provides an honest planning estimate with visible boundaries, factors and uncertainty instead of a single unsupported carbon number.

SENSORS AND TELEMETRY

Modality and location: Validated weights or allocations, treatment destinations, collection distance or route evidence at approved aggregate level, vehicle and fuel class where available and current approved emission factors; no continuous driver or vehicle tracking is required. Coverage, timing and aggregation: Cover the selected streams, collection stages and treatment boundary for the stated period. Aggregate transport and treatment components separately by evidence class and retain unknown mass or destination. Measurement discipline: Use factor units and versions consistently, avoid double counting transfer stages, validate distance sources and apply uncertainty ranges for allocation, vehicle load share and treatment factors. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: Sustainability owns the accounting boundary and factor choice; the contractor owns transport and destination evidence; the Estates waste lead owns mass and site scope; Finance may own activity records. Use current authorised methodology, source evidence and factor versions. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The estimate is valid only for the named boundary, period and factor set. Direct activity evidence outranks generic assumptions. Conflicting destination or transport evidence creates scenarios or unknown components, not a preferred single value.

ANALYSIS REQUIRED

Operation: calculation, estimate and scenario comparison. Decision interval: quarterly, annually and before an authorised option appraisal. Minimum evidence: defined boundary, valid mass or allocation, supported treatment route, transport basis where included and current approved factor with uncertainty. Method: calculate transport and treatment ranges, reconcile units and mass, present low, central and high scenarios and identify dominant uncertainty. Uncertainty: show excluded lifecycle stages, factor uncertainty, unknown destination, allocation error, avoided-burden choices and non-comparable methods. Safe failure: report emissions unavailable for unsupported components and provide only a partial range with explicit coverage.

ANSWER BOUNDARY

Do not certify a carbon footprint, imply precise avoided emissions, choose favourable factors without authority, identify drivers or claim that an estimate proves environmental benefit. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

WM-080

STAKEHOLDER QUESTION

Which evidence-backed actions offer the best combined waste, cost, service and emissions benefit for the next planning cycle?

WHY THIS IS USEFUL

Gives authorised owners a transparent shortlist of practical actions with trade-offs, dependencies and evidence gaps visible.

SENSORS AND TELEMETRY

Modality and location: The verified outputs of prevention, reuse, service, cost, contamination and emissions analyses, plus current constraints, owner capacity and delivery dependencies; no new sensing is introduced at this stage. Coverage, timing and aggregation: Cover only actions with a defined scope, accountable owner and evidence from the same planning horizon. Refresh when a source figure, constraint or owner commitment changes; aggregate benefits, costs, service effects, risks, confidence and dependencies per action. Measurement discipline: Require versioned source figures and consistent units, prevent double counting across linked actions, keep measured and estimated benefits separate and test sensitivity to key assumptions. Use traceable calibration for measured quantities or a versioned functional check for discrete devices. Align device and gateway clocks to the approved time source and retain source, receipt and processing times. Report uptime, gaps, resets and drift against a reference; preserve device or check ID, mapping version, firmware or checklist version and provenance. Treat evidence as unavailable when deployment, mapping, permission, commissioning or current health is not verified; never fill gaps silently.

AUTHORITATIVE SUPPORTING DATA

Owners and sources: The Estates waste lead coordinates options; Sustainability owns environmental criteria; Finance owns financial evidence; Procurement and Contract Management own supplier actions; safety, accessibility, security and specialist owners control constraints. Use current approved source analyses and owner inputs. Precedence and time: current emergency direction and competent-person decisions come first; then current approved procedures, contracts, manifests and owner records; then verified current observation; telemetry, calculations and estimates come last. Retain effective, observed and retrieved times, record IDs, versions, role permissions and transformation lineage. Freshness and conflict: The shortlist is advisory and expires when evidence or constraints change. Competent and accountable owner decisions outrank scoring. Actions with blocking safety, permission or evidence gaps cannot be ranked as ready.

ANALYSIS REQUIRED

Operation: comparison and multi-criteria recommendation. Decision interval: for each agreed planning cycle and after a material evidence change. Minimum evidence: defined candidate actions, traceable benefit and cost ranges, current constraints, accountable owners, dependencies and a transparent decision rule. Method: normalise only commensurable measures, present a Pareto-style shortlist and scenario rankings, expose trade-offs and identify the smallest next evidence or implementation decision. Uncertainty: show confidence by criterion, weight sensitivity, correlated benefits, double-count risk, delivery uncertainty and actions excluded for missing evidence. Safe failure: provide no overall winner; return the supported trade-off set and the exact owner decisions or evidence gaps required next.

ANSWER BOUNDARY

Do not authorise projects, spending, procurement, staffing or contract changes, hide adverse effects, promise combined benefits or turn uncertain estimates into a definitive ranking. The service must not identify or blame people, inspect confidential contents, classify unknown hazardous material, authorise handling or transport, certify compliance, override current procedures, manifests, contractors or competent persons, or control bins, compactors, lifts, doors, gates or alarms.

Decision coverage and supported response states

1

bin assets, locations and waste streams
Questions 01-10 | 10 distinct decisions

2

fill levels, collection timing and service demand
Questions 11-20 | 10 distinct decisions

3

segregation, contamination and signage
Questions 21-30 | 10 distinct decisions

4

hazardous, confidential and specialist waste
Questions 31-40 | 10 distinct decisions

5

routes, lifts, loading areas and access
Questions 41-50 | 10 distinct decisions

6

contractor coordination, storage and collections
Questions 51-60 | 10 distinct decisions

7

weights, costs, provenance and reporting
Questions 61-70 | 10 distinct decisions

8

prevention, reuse, recycling and emissions reduction
Questions 71-80 | 10 distinct decisions

SUPPORTED RESPONSE STATES

- Supported** Minimum evidence is met; authority, timing and provenance are current.
- Limited** A bounded answer is possible, but coverage or precision is explicitly constrained.
- Conflicting** Authoritative sources disagree; values remain separate and the owner resolves them.
- Stale** The last valid evidence is older than the decision-specific freshness limit.
- Unavailable** Deployment, mapping, permission, health or minimum evidence is not verified.
- Restricted** An answer exists but details are withheld from the requesting role.

DECISION CHAIN

- 1. Define** Question, owner, decision interval and current rule.
- 2. Verify** Deployment, mapping, permission, health and source identity.
- 3. Align** Effective, observed, retrieved and processing times.
- 4. Analyse** Named operation, minimum evidence, method and uncertainty.
- 5. Bound** Prohibited inference, disclosure, authorisation and control.
- 6. Act** A concrete owner action, or a safe failure with the next evidence step.

Evidence, authority and permissions rules

1. PROPOSED MEANS PROPOSED

No sensor, integration, mapping or analytic is treated as an Abacws capability until deployment, commissioning, location and asset mapping, permission and current health have been independently verified.

2. CURRENT AUTHORITY WINS

Emergency direction and competent decisions come first, followed by current approved procedures, contracts, manifests and owner records. Verified observation follows. Telemetry, calculations and estimates are supporting evidence only.

3. TIMES STAY DISTINCT

Preserve effective time, observed time, source time, receipt time, processing time and retrieval time. Align clocks where possible, expose offsets and never reorder conflicting events silently.

4. MEASUREMENT LIMITS ARE VISIBLE

State modality and placement, coverage, sampling and aggregation, calibration or functional checks, uptime, gaps, resets, drift, mapping version and provenance. Missing evidence remains missing.

5. CONFLICTS ARE NOT AVERAGED AWAY

Show each conflicting source with owner, time and version. Apply explicit precedence only where authorised; otherwise return conflicting and assign resolution to the current source owners.

6. PERMISSIONS SHAPE THE ANSWER

Use role-level access and least disclosure. A restricted answer can expose ready, not-ready or contact-the-owner without revealing confidential contents, store locations, commercial terms or individual access histories.

7. SAFE FAILURE IS AN OUTPUT

When minimum evidence is not met, preserve the current authorised procedure, exclude unsupported values, request the smallest necessary check and return limited, stale, conflicting, unavailable or restricted.

8. ANALYTICS NEVER OPERATE EQUIPMENT

The service cannot unlock doors, call lifts, open gates, reset alarms, start compactors, empty bins, alter routes, book transport, approve spending or change records at source.

Operational readiness and safe response contract

BEFORE USING ANY ANSWER

- 1 **Identity** Is the asset, station, route, store, collection, document or figure uniquely identified?
- 2 **Authority** Is the current owner and effective procedure, contract, manifest or competent direction known?
- 3 **Deployment** Is the proposed evidence actually deployed, mapped, commissioned, permitted and healthy now?
- 4 **Coverage** Does spatial and temporal coverage match the decision, with sampling and aggregation stated?
- 5 **Quality** Are calibration or functional checks current, clocks aligned and uptime, gaps, resets and drift visible?
- 6 **Traceability** Can each value be traced to original records, versions, times, permissions and transformations?
- 7 **Minimum evidence** Does the record-specific gate pass without unsafe observation or unsupported gap filling?
- 8 **Uncertainty** Are precision, confidence, scenarios, alternatives and non-comparable periods explicit?
- 9 **Boundary** Is the answer free from person identification, content inspection, specialist classification, certification, authorisation and control?
- 10 **Action** Does the output name a concrete authorised next step or a conservative safe failure?

SAFE RESPONSE CONTRACT

SUPPORTED ANSWER

State the interval, owners, evidence times, operation, result, uncertainty and authorised action.

LIMITED ANSWER

State the supported subset, omitted scope and why the limitation matters.

CONFLICTING ANSWER

Show each source and precedence; identify the owner who must resolve the conflict.

STALE ANSWER

Give the last valid time, why it is too old and the smallest current check.

UNAVAILABLE ANSWER

Name the failed deployment, mapping, permission, health or evidence gate. Do not substitute assumption.

RESTRICTED ANSWER

Expose only what the role needs and withhold sensitive detail.

ALWAYS PROHIBITED

No person identification or blame; no confidential-content inspection; no classification of unknown hazardous material; no handling or transport authority; no compliance or safety certification; no override of current procedures, manifests, contractors or competent persons; no physical equipment control.

A useful answer is the smallest current, traceable and permitted answer that supports the authorised waste-management decision without inventing certainty or controlling the building.